Financial Frictions, Market Power, and Innovation

Pedro Armada*

June 2026

Latest version

Abstract

This paper investigates how financial frictions and market power interact in shaping firms' incentives to innovate. Using administrative data covering the near-universe of Portuguese firms, I document that innovative firms are larger, more productive, charge higher markups, and rely more on equity. I also show that firms with higher markups accumulate equity faster. Motivated by the empirical evidence, I augment a general equilibrium framework of heterogeneous producers with imperfect competition and innovative technology. Using the calibrated model, I show that the financing role of market power is quantitatively important and becomes stronger when firms face tighter borrowing constraints. These findings suggest that market power can act as an imperfect substitute for missing financial markets, underscoring the importance of tailoring competition policy to financial development.

JEL codes: E2, E6, D4, O16, O3, L1

Keywords: innovation, R&D, markups, market power, productivity, financial frictions

*Halle Institute for Economic Research and Friedrich Schiller University Jena. Email: pedro.armada@iwh-halle.de. Website: www.pedroarmada.com. I am indebted to my advisors, Johanna Francis, Patricia Gomez-Gonzalez, and Philip Shaw, for their guidance and support. I also thank Dean Corbae, Utteyo Dasgupta, Jefferson Frank, Wouter den Haan, Aubhik Khan, Subha Mani, Javier Miranda, and Dennis Novy, as well as seminar and conference participants at the Friedrich Schiller University Jena, the Halle Institute for Economic Research, Queen Mary University of London, the Bank of Lithuania, the Bank of Spain, the Eastern Economic Association Annual Meeting, the European Economic Association Meeting, the Midwest Macroeconomics Meeting, the North American Summer Meeting of the Econometric Society, and the Southern Economic Association Annual Meeting for helpful comments and discussions. I thank the Banco de Portugal Microdata Research Laboratory (BPLIM) team for their help with the data. All outputs derived from confidential microdata were subject to disclosure review and approved for release in accordance with BPLIM's disclosure-control procedures. Any errors are my own.

1 Introduction

The relationship between market power and innovation is often understood as a question of *incentives*. Competition may increase innovation through the replacement or escape-competition effects (Arrow, 1962; Aghion et al., 2005), while market power may raise innovation incentives by increasing appropriability or allowing incumbents to protect rents (Schumpeter, 1942; Gilbert and Newbery, 1982; Romer, 1990; Aghion and Howitt, 1992).

This paper builds on this view by adding a *financing* dimension to this relationship. When external finance is limited, markups allow firms to generate and retain internal funds, improving their ability to finance innovation over time. This financing channel is crucial because innovation investments are risky, uncertain, and often embodied in intangible assets that are difficult to pledge as collateral. As a result, higher markups can help firms accumulate internal resources, and relax the financing constraints needed to sustain innovation. Market power therefore affects innovation dynamically through two complementary channels: by increasing the payoff to successful innovation and by strengthening the financial capacity required to sustain innovation over time.

I develop this argument in two parts. First, I provide empirical evidence on the joint relationship between innovation, financing structure, and market power using administrative data covering the near-universe of Portuguese firms. Portugal provides a useful setting for this analysis because its bank-dependent financial system makes firms' balance sheets especially important for investment and innovation. The data combine detailed balance-sheet information with R&D employment, allowing me to study how innovation activity is related to firm-level productivity, markups, and the composition of financing between debt and equity. Second, I develop a quantitative macroeconomic model with heterogeneous firms, endogenous innovation, financial frictions, and variable markups. The model rationalizes the empirical facts and provides a laboratory to quantify the importance of the financing channel of market power for innovation and aggregate outcomes.

The first set of empirical results documents that innovation is concentrated among large firms. I measure innovation using firms' R&D employment and rank firms by their size relative to other firms in the same industry. Innovation rises sharply across the relative size distribution: firms in the upper tail are substantially more likely to employ R&D workers and allocate a larger share of their workforce to R&D. Using production-function methods to estimate firm-level productivity and markups, I then show that firms with greater R&D employment exhibit higher productivity and higher markups than otherwise comparable firms. These results are robust to controlling for overhead expenditures, suggesting that higher measured markups among innovative firms are not driven by higher fixed operating costs associated with innovation. Together, the productivity and markup estimates imply incomplete pass-through: innovation-related efficiency gains are not fully passed on to consumers through lower prices, but are instead partly retained by firms through higher markups.

The second set of results links innovation to firms' financing structure. Innovative firms rely less on debt and hold more equity than otherwise similar firms. These differences remain after controlling for liquidity holdings, mitigating the concern that innovative firms have less need for debt because they hold larger cash buffers. They also persist after controlling for asset collateralizability, suggesting that the lower leverage of innovative firms is not explained by differences in the tangibility of their balance sheets. Moreover, the lower leverage of innovative firms is present across debt maturities, with especially pronounced differences in long-term borrowing. These patterns are consistent with the view that innovation is associated with lower debt capacity and a stronger reliance on internal finance.

I then examine the role of internal financial capacity for innovation. Firms with larger equity positions are more likely to employ R&D workers and devote a larger share of employment to R&D activities. This relationship holds both at the extensive margin of whether the firm innovates and at the intensive margin of how much labor it allocates to innovation. The results remain positive and statistically significant after controlling for leverage and collateral ratios. Thus, even among firms with similar debt positions and asset tangibility, those with larger equity positions are more likely to innovate. This pattern is consistent with internal funds playing an important role in supporting innovation when external finance cannot fully substitute for firms' own resources.

Finally, I provide evidence for the financing channel of market power by connecting market power to the accumulation of internal funds. I find that firms with higher markups experience faster equity growth, both in the full sample and among firms that employ R&D workers. I then conduct several additional checks to address alternative interpretations. First, high-markup firms may accumulate equity faster simply because they invest less. I show that the relationship is not driven by lower investment rates. Second, markups and equity growth may be jointly shaped by sector-specific trends or demand shocks. The result remains positive after including industry-by-year fixed effects, which absorb shocks common to firms in the same industry and year. Third, the estimates may reflect persistent differences across firms, such as managerial quality. The relationship also survives the inclusion of firm fixed effects, which exploit within-firm variation over time. This evidence supports the paper's central mechanism: market power helps firms accumulate internal financial resources, thereby relaxing financing constraints and supporting innovation over time.

Motivated by these empirical findings, I develop a quantitative general equilibrium model with heterogeneous firms. In the model, firms choose production inputs, financing, and R&D investment. Innovation raises productivity, but borrowing constraints limit firms' ability to finance innovation externally, making internal funds an important determinant of innovation activity. Firms compete under non-CES demand, which generates variable markups and incomplete pass-through of productivity gains into prices. As firms become more productive, they expand their market shares and charge higher markups. Higher markups increase profits and accelerate the accumulation of equity, which relaxes future financing constraints. The model therefore captures a dynamic financing channel of market power. Beyond affecting the incentives to innovate, market power influences firms' ability to self-finance innovation and growth over time.

The calibration strategy targets the forces that are central to the mechanism: the scaling and concentration of innovation activity, the markup distribution, and the tightness of financial constraints. The quantified model also performs well along several untargeted dimensions, including the concentration of economic activity among large firms, the aggregate importance of innovative firms, and the size gradient in innovation activity.

I then use the calibrated model to quantify the financing role of markup rents. I introduce an internal-finance wedge that restricts the extent to which markup rents can be carried forward as firm equity. The wedge allows me to separate the financing channel from the incentive channel. In one set of counterfactuals, firms' choices of technology, factor inputs, and prices are held fixed, so the wedge affects only the accumulation of internal funds. In a second set, firms also internalize the wedge when making operating decisions. This decomposition shows that markup rents affect innovation not only by increasing the current return to innovation, but also by relaxing future financing constraints. Even when operating incentives are held fixed, restricting retained markup rents lowers aggregate innovation and output. For instance, when firms are prevented from retaining 10% of markup rents, R&D participation falls by about 9% and output falls by about 3.3%. Relative to the full response, the financing channel accounts for about 11% of the decline in R&D participation and roughly 31% of the output decline in partial equilibrium.

General-equilibrium responses partly dampen the wedge effects, but unevenly across channels. The wedge reduces firms' ability to accumulate equity, lowering firm scale and placing downward pressure on wages. At the same time, lower aggregate equity raises the equilibrium interest rate. These price adjustments attenuate the incentive channel more than the financing channel. Lower wages partially offset the contraction in firms' production and innovation incentives, but the higher return on retained funds is insufficient to fully compensate for the equity that firms fail to accumulate when markup rents cannot be retained. As a result, the financing channel remains sizable even after equilibrium prices adjust. Comparing these responses reveals that a nontrivial fraction of the total effect operates through internal finance, a mechanism that is absent from models in which market power influences innovation only through current incentives.

I also examine whether restricting the accumulation of markup rents materially reduces market power itself. The results suggest that the answer is largely no. When firms are prevented from retaining 10% of markup rents, aggregate markup rents fall by 2.6% to 3.1% when only the financing channel is active, and by 7.9% in partial equilibrium when firms also adjust operating decisions. By contrast, the revenue-weighted average markup falls by at most 1.0%. Thus, limiting the accumulation of markup rents reduces the flow of rents available for internal finance, but does not substantially compress the markups charged by firms.

Finally, I show that the importance of the markup-rent financing channel depends on the severity of borrowing constraints. When borrowing constraints are tighter, firms rely more heavily on retained earnings, and the internal-finance wedge has larger effects on aggregate innovation and output. As external finance becomes more accessible, firms can substitute borrowing for retained markup rents, and the response to the wedge becomes weaker. This pattern further validates the mechanism: the importance of markup rents declines when firms are less dependent on internal funds. The results support the interpretation of market power as an imperfect substitute for missing financial markets.

These findings have important policy implications. Competition policies that reduce the rents generated by market power may unintentionally weaken innovation when firms face financial frictions. Because markup rents provide internal funds that help finance future R&D, limiting firms' ability to retain those rents can reduce innovation by tightening internal financing capacity. In financially constrained economies, market power can act as a partial substitute for missing financial markets. But it relies on distortions that drive a wedge between prices and marginal costs, leading to inefficient resource allocation. Policies that improve access to external finance can reduce firms' dependence on retained markup rents, allowing innovation to be financed without relying as heavily on market power.

This paper contributes to several strands of literature. First, it contributes to the literature on competition and innovation. Canonical theories emphasize that market structure determine the payoff to innovation through appropriability, replacement effects, preemption, and escape-competition forces (Schumpeter, 1942; Arrow, 1962; Gilbert and Newbery, 1982; Romer, 1990; Aghion and Howitt, 1992; Aghion et al., 2005). I complement this view by showing that market power also affects firms' ability to finance innovation. In doing so, the paper also adds to recent work on rents and R&D misallocation (Aghion et al., 2023, 2025), by emphasizing that rents affect innovation not only through the allocation of incentives across firms, but also through the allocation of financial capacity. Second, the paper contributes to the literature on financial frictions for innovation and intangible investments. A large body of work shows that innovation is difficult to finance externally (Hall and Lerner, 2010; Brown et al., 2009; Nanda and Kerr, 2015) and often embodied in intangible assets that are hard to pledge as collateral (Crouzet and Eberly, 2018, 2019; Crouzet et al., 2022; Peters and Taylor, 2017). This paper adds to this literature by studying how firms generate the internal funds needed to sustain innovation when external finance is limited. Third, the paper relates to the literature on market power and aggregate outcomes. Recent work documents rising markups and studies their implications for allocative efficiency, labor shares, business dynamism, and aggregate productivity (De Loecker et al., 2020; Autor et al., 2020; Edmond et al., 2023). This paper contributes by emphasizing a distinct dynamic channel through which markups affect aggregate productivity: how markups influence firms' ability to self-finance innovation. Lastly, the paper contributes to quantitative models of heterogeneous firms with financial frictions and aggregate productivity. Existing work shows that financial frictions distort firm dynamics, capital accumulation, and aggregate productivity by limiting the growth of constrained firms (Buera et al., 2011; Midrigan and Xu, 2014; Moll, 2014). This paper extends this framework by combining financial frictions with endogenous innovation and variable markups. In the model, financial frictions distort not only physical capital accumulation, but also the allocation of R&D and the evolution of firm productivity.

The rest of the paper is organized as follows. Section 2 describes the firm-level data, the construction of the main variables, and the estimation of firm-level productivity and markups. Section 3 presents the empirical evidence linking innovation activity, financing structure, and market power. Section 4 develops a quantitative model with heterogeneous firms, endogenous innovation, financial frictions, and variable markups. Section 5 calibrates the model and uses it to quantify the financing channel of market power and study counterfactual economies. Section 6 concludes.

2 Data

This section presents the firm-level data used in the paper. I first introduce the data source and define the main variables used in the analysis. I then describe the sample selection and cleaning procedures applied to the data. I report summary statistics and assess the coverage and representativeness of the R&D employment measures by comparing them with official statistics. Finally, I outline the estimation procedure used to recover firm-level productivity and markups.

2.1 Data Source

The main data source used in this paper is the Central Balance Sheet Harmonized Panel Data (CBHP), provided by Banco de Portugal Microdata Research Laboratory (BPLIM) (2025). The CBHP is a longitudinal dataset containing annual economic and financial information on Portuguese firms over the period 2006–2023. The dataset covers the universe of Portuguese non-financial corporations. This includes all market producers whose primary activity is the production of non-financial goods and services, encompassing private and public corporations, cooperatives, partnerships, and other legally recognized entities. The underlying data are collected through the *Informação Empresarial Simplificada* (IES), a mandatory annual reporting system submitted electronically by firms to the Portuguese Ministry of Finance. The unit of analysis is the firm, identified through an anonymized tax identifier. In the case of multi-establishment firms, the data reflect consolidated accounting information at the firm level.

A key feature of the dataset is the harmonization of variables over time. In 2010, Portugal introduced substantial changes in accounting standards. The CBHP addresses this break by constructing harmonized variables that ensure comparability over time. This harmonization is implemented by mapping variables across accounting systems and, when necessary, aggregating or reconstructing measures to maintain consistency. Moreover, the dataset is subject to quality control procedures conducted by Banco de Portugal.

2.2 Variable Definitions

The main variables used throughout the empirical analysis are defined as follows. Output is measured by total turnover, which corresponds to sales of goods and services. Capital is fixed assets, which include both tangible and intangible assets. Labor is measured using employee expenses, including wages and other labor-related costs. Materials are defined as the sum of the cost of goods sold and materials consumed plus supplies and external services.[1] Employment corresponds to the total number of employees. R&D employment measures the number of employees allocated to research and development activities.[2] Debt corresponds to total financial debt and equity refers to total shareholder equity. All financial variables are converted into real 2023 euros using the Portuguese GDP deflator obtained from the annual national accounts published by Statistics Portugal (INE).

Industries are classified according to the Portuguese Classification of Economic Activities (CAE Rev. 3), as recorded in the Central Registry of Companies. CAE Rev. 3 corresponds to the Portuguese implementation of NACE Rev. 2 and is therefore directly comparable to European industry classifications. The main industry is defined primarily using gross value added at factor cost and, when unavailable, using turnover or employment information. The classification is available at up to the 5-digit industry level, and BPLIM provides harmonized industry codes that remain consistent throughout the sample period.

Firm age is the difference between the reference year of the balance sheet and the year of incorporation plus one. The year of incorporation corresponds to the year the firm began operations and is complemented using information from the Central Registry of Companies. Exporter is an indicator equal to one if the firm reports positive sales or service exports to foreign markets. Single-owner is an indicator equal to one for firms classified as *Sociedades Unipessoais por Quotas* (single-holder private limited liability companies). Multi-establishment is an indicator equal to one for firms operating more than one establishment. The underlying establishment information includes both domestic and foreign establishments, where establishments are defined as geographically identified units carrying out economic activity with one or more workers. Legal form is a categorical variable indicating the firm's legal status, distinguishing between private limited liability and public limited liability companies.

[1] In Portuguese accounting standards, the cost of goods sold and materials consumed reflects direct costs incorporated into production or resale activities, whereas supplies and external services correspond to purchased services and external operating expenditures that are not physically embodied in the final good.

[2] According to Portuguese accounting standards, firms must identify employees directly involved in R&D activities. These typically include scientists, engineers, technicians, and other staff involved in studies of design, manufacturing, and commercialization of new products, studies of commercialization or industrial rationalization, etc.

I also construct several financial ratios used throughout the analysis. The leverage ratio is defined as total debt relative to total assets. The equity ratio is total equity relative to total assets. The collateral ratio is measured as tangible fixed assets relative to total assets. The cash ratio is defined as cash and cash equivalents relative to total assets. The overhead ratio is defined as supplies and external services relative to total expenses. The cost of debt is measured as interest expenses relative to total debt. Finally, the share of long-term debt is defined as non-current financial debt, corresponding to financial liabilities due after one year, relative to total financial debt.

2.3 Sample Selection

I next describe the construction of the firm-level sample used in the empirical analysis. I begin by restricting the sample to firms organized as private and public limited liability companies, which in Portuguese law correspond to *sociedades por quotas* and *sociedades anónimas*, respectively. These legal forms account for the vast majority of corporate activity in Portugal and are characterized by limited liability for shareholders. I exclude firms with missing, invalid, or unspecified legal form classifications. I further restrict the sample to domestically owned firms operating in the private sector. In particular, I exclude branches of foreign firms using a flag constructed by Banco de Portugal based on information contained in firms' legal names, such as the expressions *sucursal* or *representação permanente*. I additionally use the institutional sector classification to exclude holding companies, foreign-controlled corporations, and firms belonging to the central, regional, or local government sectors.

I next retain only firms that are actively operating at the time of reporting. To do so, I combine several measures of firm status available in the data, including the reported status at the end of the fiscal year, a liquidation indicator constructed by Banco de Portugal based on the expression *em liquidação* appearing in firms' legal names, and the indicator of economic activity. I keep only firms classified as actively operating and exclude firms that are dissolved, liquidated, suspended, awaiting the start of activity, ceased, or with unknown status. In addition, I require the reported information to refer to a period of at least 360 days. I also restrict the sample to standard annual declarations of activity, excluding declarations associated with consolidation procedures, cessation of activity, or other special reporting purposes.

Finally, I restrict the sample based on firms' location, sector of activity, and data availability. I retain only firms headquartered in mainland Portugal, excluding firms located in the autonomous regions of Madeira and the Azores as well as firms with missing or unspecified district information. I further exclude firms without information on their main sector of activity and firms for which the main activity accounts for less than 75% of total turnover. The main sector is defined using the Portuguese Classification of Economic Activities (CAE Rev. 3), primarily based on gross value added at factor cost and, when unavailable, using turnover or employment information. I also exclude firms operating in financial and insurance activities, public administration and defense, activities of households as employers, and activities of extraterritorial organizations and bodies. Finally, I drop firms with missing information on the year of incorporation.

2.4 Data Cleaning

This section describes the data cleaning procedures and consistency checks applied to the firm-level data. The cleaning strategy follows standard approaches in the literature, including Gopinath et al. (2017) and Kalemli-Özcan et al. (2024), adapted to the features of the CBHP data. Compared with commercial databases such as Orbis, the CBHP data exhibit lower levels of missingness and benefit from harmonization and validation procedures conducted by Banco de Portugal. As a result, the additional cleaning steps implemented here are comparatively limited and intended to remove economically implausible observations and outliers that could distort the estimated production functions.

I begin by removing firm-year observations with zero or negative values for sales, materials, capital, labor, and equity. Next, I exclude firms that report negative total assets, debt, or liabilities in any year. I also drop firm-year observations in which debt exceeds total liabilities, as well as observations in which cash holdings or fixed assets exceed total assets. In addition, I exclude observations where labor or materials exceed sales, and observations where interest expenses exceed total debt. To reduce the influence of outliers, I construct ratios of capital, labor, and materials relative to sales, as well as cash holdings, fixed assets, and debt relative to total assets. For each ratio, I compute the 0.1 and 99.9 percentiles within 2-digit industries and exclude observations outside these bounds.

Finally, I winsorize all continuous variables at the 1st and 99th percentiles. This includes variables measured in levels, such as output, capital, labor, debt, and equity, as well as financial ratios including the leverage ratio, equity ratio, collateral ratio, cash ratio, overhead ratio, the cost of debt, and the share of long-term debt. I also winsorize the estimated markups, productivity measures, the first difference of log equity, and the first difference of log capital.

2.5 Summary Statistics

Table 1 and Table 2 report summary statistics for the main firm-level variables and employment measures used in the analysis. The average firm in the sample generates approximately 1.1 million euros in output, spends 752 thousand euros on materials and 193 thousand euros on labor, holds 329 thousand euros in capital, and has equity of roughly 467 thousand euros. The average leverage ratio is 22 percent, while the average equity ratio is 42 percent. Roughly 23 percent of firms export and 13 percent operate multiple establishments. Innovative firms are substantially larger than the average firm in the sample, employing 120 workers on average, compared with 12 across all firms.

Among innovative firms, the average number of R&D workers is approximately 9.

Table 1. Summary statistics

	Mean	SD	p10	p25	p50	p75	p90
Output	1,121	3,009	53	108	260	739	2,233
Capital	329	1,019	2	10	42	178	644
Labor	193	450	12	25	59	151	399
Materials	752	2,035	21	53	153	484	1,574
Debt	296	952	0	2	35	152	561
Equity	467	1,352	11	30	91	283	906
Age	15.92	12.99	3	7	13	22	32
Exporter	0.23	0.42	0	0	0	0	1
Single-owner	0.23	0.42	0	0	0	0	1
Multi-establishment	0.13	0.34	0	0	0	0	1
Leverage ratio	0.22	0.22	0.00	0.00	0.16	0.35	0.54
Equity ratio	0.42	0.27	0.09	0.20	0.39	0.63	0.83
Collateral ratio	0.26	0.24	0.01	0.06	0.18	0.40	0.64
Cash ratio	0.20	0.22	0.01	0.03	0.11	0.30	0.55
Overhead ratio	0.28	0.20	0.07	0.13	0.22	0.40	0.59

Notes: The table reports summary statistics for the main firm-level variables used in the analysis. Output, capital, labor, materials, debt, and equity are expressed in thousands of real 2023 euros. Output is measured using sales, capital is fixed assets, labor refers to employee expenses, and materials are defined as the sum of cost of goods sold plus supplies and external services. Exporter is an indicator equal to one for exporting firms. Single-owner is an indicator equal to one for firms with a single shareholder. Multi-establishment is an indicator equal to one for firms operating more than one establishment. The leverage ratio is defined as debt over total assets, the equity ratio as equity over total assets, the collateral ratio as tangible fixed assets over total assets, the cash ratio as cash and cash equivalents over total assets, and the overhead ratio as supplies and external services divided by total expenses. The sample contains 2,631,219 firm-year observations.

2.6 Coverage and Representativeness

The data used in this paper are drawn from the *Informação Empresarial Simplificada* (IES), a mandatory administrative reporting system covering the near-universe of firms operating in Portugal. The IES forms the basis for several official statistical products produced by Portuguese statistical authorities, including Banco de Portugal and Statistics Portugal (INE). As a result, the financial information contained in the data is comprehensive. To evaluate the coverage of R&D employment, Table 3 compares the number of R&D employees observed in the data to official statistics from Eurostat on business enterprise R&D personnel in Portugal. There are important differences in how R&D employment is defined in the sample and the official statistics. First,

Table 2. Employment composition by innovation status

	Variable	Mean	SD	p10	p25	p50	p75	p90
All firms	Employment	12.28	120.35	1	2	4	8	19
	R&D employment	0.15	3.81	0	0	0	0	0
	Non-R&D employment	11.58	96.57	1	2	4	8	18
Innovative firms	Employment	120.28	369.67	2	8	37	113	266
	R&D employment	8.98	27.64	1	1	3	8	19
	Non-R&D employment	111.30	360.51	0	3	30	104	251

Notes: The table reports summary statistics for employment measures across all firms and innovative firms. Innovative firms are defined as firms with positive R&D employment. Employment corresponds to the total number of workers. R&D employment refers to workers engaged in R&D activities, while non-R&D employment is defined as total employment net of R&D workers. The sample contains 2,629,933 firm-year observations for total employment and 1,074,552 firm-year observations for the decomposition between R&D and non-R&D employment. The subsample of innovative firms contains 18,527 firm-year observations.

the Eurostat figures are constructed using a broader definition of R&D employment that includes researchers, technicians, managers, administrative personnel, and external contributors supporting R&D activities. Second, the official statistics include sectors and institutional units that are excluded from the present analysis, including financial firms and other entities outside the population of non-financial corporations considered here. Third, the data cleaning procedures required for production function estimation impose additional sample restrictions on the accounting data, which further reduces measured coverage. Thus, the reported coverage ratios should be interpreted as conservative estimates of the representativeness of the sample.

Despite these considerations, the coverage ratios remain substantial, particularly in the sectors most relevant for innovation activity. Using headcounts, the sample captures on average 24% of total R&D employment over the sample period, with substantially higher coverage in manufacturing (32%) and trade (45%). When measured in full-time equivalents (FTE), overall coverage rises to 46%, reaching 65% in manufacturing and 78% in trade. Coverage in ICT is also sizable, amounting to 24% in headcounts and 40% in FTE terms. The firm-size breakdown further suggests that the sample captures a substantial share of R&D activity across the size distribution, with coverage rates between 39% and 54%.

Table 4 further compares the composition of R&D employment in the sample to the official statistics across sectors and firm-size categories. Across sectors, the sample exhibits a somewhat larger concentration of R&D employment in manufacturing and a smaller concentration in professional services relative to the official statistics, though it aligns well with the observed shares in ICT and trade. Moreover, the firm-size distribution of R&D employment closely matches that observed in the official data.

Table 3. Coverage of R&D employment

Panel A: By firm size

	FTE
0–9	0.54
10–49	0.47
50–249	0.52
250–499	0.45
500+	0.39
Total	0.46

Panel B: By sector

	Headcounts	*FTE*
Manufacturing	0.32	0.65
ICT	0.24	0.40
Professional	0.11	0.18
Trade	0.45	0.78
Other	0.11	0.30
Total	0.24	0.46

Notes: The table reports average coverage ratios of R&D personnel in the sample relative to official statistics from Eurostat for the Portuguese business enterprise sector over the 2006–2023 period. Sectoral coverage ratios are reported using both headcounts and full-time-equivalent (FTE) measures. Firm-size coverage ratios are available only for FTE measures, since Eurostat does not provide size-class breakdowns for headcounts. Sector groups correspond to NACE Rev. 2 sections: Manufacturing (C), Wholesale and retail trade; repair of motor vehicles and motorcycles (G), Information and communication (J), Professional, scientific and technical activities (M), and a residual category including all remaining sectors. Firm size classes correspond to firms with 0–9, 10–49, 50–249, 250–499, and 500+ employees.

Table 4. Composition of R&D employment

Panel A: By firm size

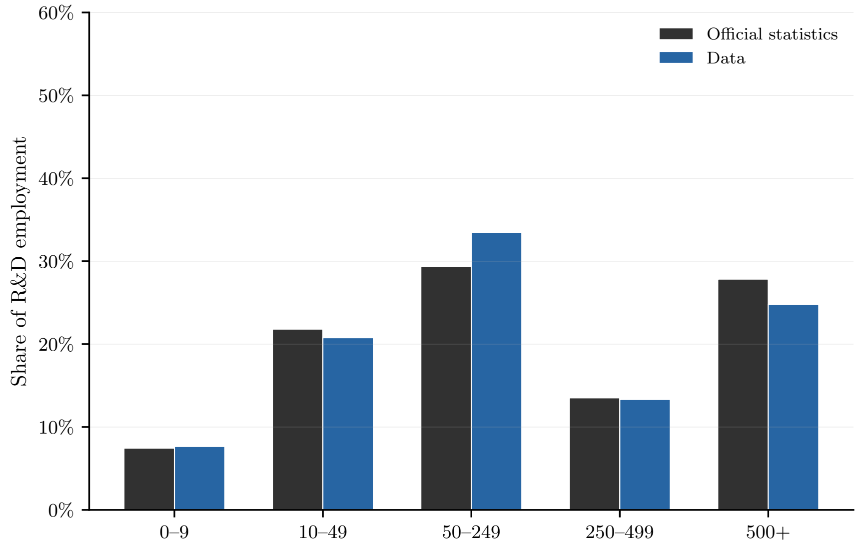
	FTE	
	Official	Sample
0–9	0.07	0.08
10–49	0.22	0.21
50–249	0.29	0.33
250–499	0.14	0.13
500+	0.28	0.25

Panel B: By sector

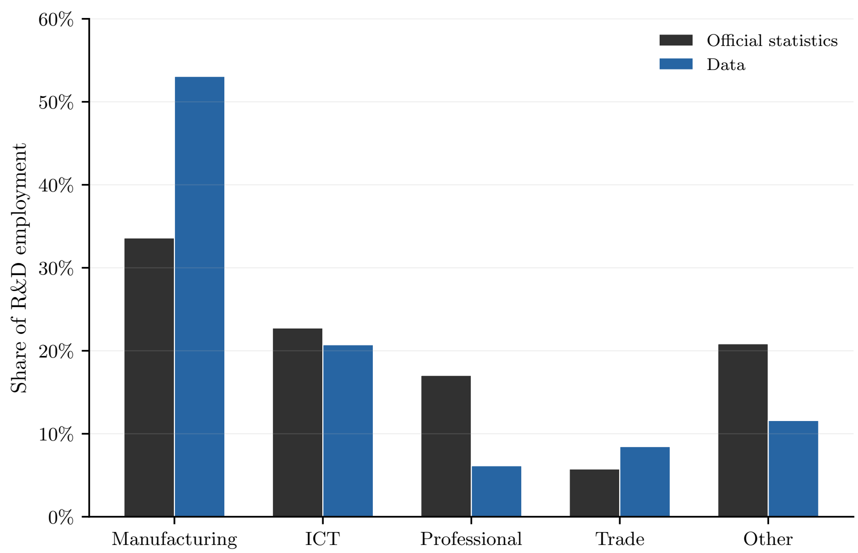
	Headcounts		*FTE*	
	Official	Sample	Official	Sample
Manufacturing	0.34	0.53	0.34	0.53
ICT	0.20	0.21	0.23	0.21
Professional	0.14	0.06	0.17	0.06
Trade	0.05	0.08	0.06	0.08
Other	0.26	0.12	0.21	0.12

Notes: The table reports the composition of R&D employment in the sample and in the official Eurostat statistics for the Portuguese business enterprise sector over the 2006–2023 period. Sectoral shares are reported using both headcounts and full-time-equivalent (FTE) measures. Firm-size shares are available only for FTE measures in the Eurostat data. Sector groups correspond to NACE Rev. 2 sections: Manufacturing (C), Wholesale and retail trade; repair of motor vehicles and motorcycles (G), Information and communication (J), Professional, scientific and technical activities (M), and a residual category including all remaining sectors. Firm size classes correspond to firms with 0–9, 10–49, 50–249, 250–499, and 500+ employees.

Figure 1. Composition of R&D Employment



(a) By firm size



(b) By sector

Notes: The figure compares the composition of R&D employment in the sample and in official Eurostat statistics for the Portuguese business enterprise sector over the 2006–2023 period. Shares are based on full-time-equivalent (FTE) measures. Panel (a) reports the composition of R&D employment by firm size. Firm size classes correspond to firms with 0–9, 10–49, 50–249, 250–499, and 500+ employees. Panel (b) reports the composition of R&D employment by sector. Sector groups correspond to NACE Rev. 2 sections: Manufacturing (C), Wholesale and retail trade; repair of motor vehicles and motorcycles (G), Information and communication (J), Professional, scientific and technical activities (M), and a residual category including all remaining sectors. Both the sample and the official statistics include all sectors of economic activity.

2.7 Production Function Estimation

This section describes the procedure used to estimate firm-level markups and productivity following the production-based approach of Hall (1988) and De Loecker and Warzynski (2012). The methodology combines production function estimation with the firm's static cost minimization problem to recover markups from accounting data. Firm production is assumed to follow

$$y_{it} = f(\ell_{it}, k_{it}) + \omega_{it} + \varepsilon_{it}, \tag{1}$$

where y_{it} denotes log output, ℓ_{it} denotes the flexible input, k_{it} denotes the predetermined input, ω_{it} is firm-level productivity observed by the firm but not by the econometrician, and ε_{it} captures measurement error and transitory shocks. In the empirical implementation, I use materials as the flexible input and capital as the predetermined input. The production function coefficients are estimated following the control function approach of Levinsohn and Petrin (2003) and Ackerberg et al. (2015). This procedure addresses the simultaneity problem arising from firms observing productivity shocks when choosing inputs. Productivity is assumed to evolve according to a first-order Markov process:

$$\omega_{it} = g(\omega_{it-1}) + \xi_{it}, \tag{2}$$

where ξ_{it} denotes the innovation to productivity. The estimation proceeds in two stages. First, an initial regression is used to obtain predicted output net of measurement error. Second, moment conditions exploiting the timing assumptions on firms' input choices are used to identify the production function coefficients while controlling for unobserved productivity. All specifications include year fixed effects and are estimated separately for each 2-digit industry. To ensure a sufficient number of observations, I restrict the estimation to 2-digit industries with at least 100 firm-year observations.

Firm-level markups are then recovered from the firm's static cost minimization problem. Under cost minimization, the markup is given by

$$\mu_{it} = \frac{\theta_{\ell,it}}{\alpha_{\ell,it}}, \tag{3}$$

where $\theta_{\ell,it}$ denotes the output elasticity of the flexible input and $\alpha_{\ell,it}$ denotes the expenditure share of the flexible input in firm revenue. The expenditure share of the flexible input is defined as $\alpha_{\ell,it} = \frac{X_{it}}{Y_{it}}$, where X_{it} denotes expenditure on the flexible input and Y_{it} denotes firm output.

Finally, firm-level productivity is obtained as the residual from the estimated production function,

$$\widehat{\omega}_{it} = y_{it} - \widehat{f}(\ell_{it}, k_{it}),$$

where $\widehat{f}(\ell_{it}, k_{it})$ denotes the estimated production function evaluated at the firm's observed inputs.

Table 5. Summary Statistics for the Production Function Estimates

	Mean	SD	p10	p25	p50	p75	p90
Returns to scale	0.92	0.08	0.82	0.89	0.93	0.95	1.02
Markups	1.73	1.07	1.01	1.13	1.37	1.86	2.84
Productivity	0.95	0.53	0.28	0.74	0.91	1.10	1.39

Notes: The table reports summary statistics for returns to scale, firm-level markups, and productivity estimates obtained using the De Loecker and Warzynski (DLW) method, with all specifications estimated separately at the 2-digit industry level using materials as the flexible input. Let $y_{it} = f(\ell_{it}, k_{it}) + \omega_{it}$ denote the production function. Returns to scale is defined as $\text{RTS}_{it} = \theta_{\ell,it} + \theta_{k,it}$, where $\theta_{j,it} \equiv \partial f(\cdot)/\partial \ln j_{it}$ for $j \in \{\ell, k\}$. Markups are computed as $\mu_{it} = \theta_{m,it}/\alpha_{m,it}$, where $\theta_{m,it}$ denotes the output elasticity of materials and $\alpha_{m,it}$ denotes the expenditure share of materials in sales. Productivity is reported in levels and corresponds to $\exp(\omega_{it})$, where ω_{it} is the estimated log productivity residual net of observed input contributions. The sample contains 1,550,506 firm-year observations.

Table 5 summarizes the distribution of estimated firm-level returns to scale, markups, and productivity. The estimates are broadly consistent with existing evidence from the production-function literature. Returns to scale are centered close to one, with a median of 0.93 and a mean of 0.92, suggesting approximately constant returns to scale on average. Markups exhibit substantial heterogeneity across firms, with a median of 1.37 and a mean of 1.73. Productivity is also highly dispersed, with a standard deviation of 0.53 in levels.

3 Empirical Analysis

This section provides the empirical foundation for the model developed in Section 4. First, I document that innovative firms are systematically larger, more productive, and charge higher markups than otherwise comparable firms, consistent with incomplete pass-through of innovation-related efficiency gains. Second, I show that innovative firms differ in their financial structure: they hold less debt, rely more on equity, and display a positive relationship between internal funding capacity and innovation activity. Third, I examine whether market power helps firms accumulate internal resources and find that that firms with higher markups experience faster equity growth.

3.1 Innovation and Firm Performance

This section documents a set of empirical patterns linking innovation activity to firm size, productivity, and market power. The analysis focuses on firms' R&D employment, which captures the allocation of labor toward innovative activities within the firm.

To examine the relationship between innovation and firm size, I rank firms according to their relative sales within each 2-digit industry-year group and then partition the distribution into equally sized ventiles. This approach allows me to account for differences in scale across industries and isolate variation in firm size within industries. Figure 2 documents a strong positive relationship between innovation activity and relative firm size, both at the extensive and intensive margins. Panel (a) plots the share of firms with positive R&D employment across the relative firm size distribution, while Panel (b) reports the average share of workers allocated to R&D activities. Larger firms are substantially more likely to engage in innovation activities. Among firms in the smallest relative size ventile, fewer than 1% report positive R&D employment, compared to roughly 16% among firms in the largest ventile. Larger firms also devote a greater fraction of their workforce to R&D activities. The average R&D employment share increases from approximately 0.4–0.6% among smaller firms to roughly 1.7% among the largest firms.

The strong size gradient in innovation activity raises the question of what drives these differences in firm size. A natural candidate is that innovation improves firm efficiency, allowing firms that engage more intensively in innovation activities to grow larger than their peers. If this is the case, firms with greater innovation activity should exhibit systematically higher productivity levels relative to otherwise similar firms operating in the same industries. To investigate the link between innovation activity and firm-level productivity, I estimate regressions of the following form:

$$\ln(\text{Productivity}_{it}) = \beta_0 + \beta_1 \ln(\text{R\&D emp}_{it}) + \Gamma' Z_{it} + \delta_s + \delta_t + \varepsilon_{it}, \tag{4}$$

where Productivity_{it} is the estimated productivity residual obtained from the production function estimation[3]. The variable $\ln(\text{R\&D emp}_{it})$ denotes the log number of workers allocated to R&D activities, Z_{it} is a vector of firm-level controls, δ_s denotes industry fixed effects, and δ_t denotes year fixed effects. The vector of controls includes firm size, age, exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. Table 6 shows that firms with higher R&D employment exhibit higher productivity levels. Quantitatively, the estimated effects are economically meaningful when evaluated over realistic changes in firms' R&D employment. Among firms with positive R&D employment, the average number of R&D workers is approximately 9 (Table 2). Increasing the number of R&D workers from 9 to 10 is therefore associated with approximately a 0.0020–0.0021 log-point increase in productivity, corresponding to about 0.20–0.21 percent higher productivity. Moreover, these differences are not explained by innovative firms simply being larger, older, more export-oriented, or disproportionately concentrated in particular regions or industries. Column (2) additionally includes overhead expenditures relative to total expenses as a control for differences in firms' operating cost structures. This is important because innovative firms may incur higher overhead expenses related to managerial, administrative, or organizational activities that accompany innovation investment. If

[3] In the absence of firm-level price data, the estimated residual corresponds more closely to revenue productivity (TFPR) rather than physical productivity (TFPQ), following the terminology in Hsieh and Klenow (2009). In the context of innovation, this measure captures not only process improvements that increase technical efficiency, but also product differentiation or quality improvements that raise revenues conditional on inputs.

these costs were correlated with measured productivity, part of the estimated relationship could mechanically reflect differences in operating structure rather than underlying efficiency. Nonetheless, the estimated coefficients remain similar after including these controls, suggesting that the productivity advantage of innovative firms is not driven by differences in cost structure.

The fact that innovative firms appear more productive does not, by itself, reveal how these productivity gains translate into market outcomes. In principle, more productive firms could pass efficiency gains entirely to consumers through lower prices. At the same time, innovation may generate differentiated products or technological advantages that allow firms to retain part of these gains in the form of greater market power. The next set of results examines whether firms with greater innovation activity also exhibit systematically higher markups. To assess the relationship between innovation and market power, I estimate the following regression:

$$\ln(\text{Markup}_{it}) = \beta_0 + \beta_1 \ln(\text{R\&D emp}_{it}) + \Gamma' Z_{it} + \delta_s + \delta_t + \varepsilon_{it}, \tag{5}$$

where $\ln(\text{Markup}_{it})$ denotes the log of firm-level markups estimated from the production function. The remaining controls and fixed effects follow the same specification as in equation (4). Table 6 reports the corresponding regression results. The estimates reveal a positive association between R&D employment and firm-level markups. In particular, a 1% increase in R&D employment is associated with approximately a 0.018% increase in firm-level markups. Evaluated at the average innovative firm, increasing R&D employment from 9 to 10 workers is associated with roughly a 0.2% increase in firm-level markups. A potential concern is that higher measured markups among R&D-intensive firms may simply reflect compensation for fixed costs associated with innovation activities, rather than differences in market power. To address this issue, column (4) again controls for overhead expenditures relative to total expenses, which proxy for fixed operating costs not directly tied to current production. Including these additional controls has only a modest effect on the estimated coefficients, which remain positive and statistically significant, suggesting that the relationship between innovation and markups is not due to higher fixed costs associated with innovation activity.

Although the absence of firm-level price data precludes a direct estimate of price responses, the joint productivity and markup estimates are informative about the pass-through of efficiency gains into prices. Using the identity $p = \mu \times MC$, changes in log prices can be decomposed as $d \ln p = d \ln \mu + d \ln MC$, implying that price responses reflect the changes in markups and marginal costs. Under a standard Cobb–Douglas production function, marginal costs are inversely proportional to firm productivity, so that $d \ln MC = -d \ln z$ holding factor prices fixed.[4] Using the estimates

[4] It is straightforward to show that $d \ln MC = -d \ln z$ under a standard Cobb–Douglas production function. Consider $y = zk^{\alpha}l^{1-\alpha}$. Cost minimization implies a marginal cost function $MC = \frac{1}{z}\left(\frac{r}{\alpha}\right)^{\alpha}\left(\frac{w}{1-\alpha}\right)^{1-\alpha}$, where w and r denote the wage and rental rate of capital, respectively. Taking logs yields $\ln MC = -\ln z + \alpha \ln r + (1-\alpha)\ln w - \alpha \ln \alpha - (1-\alpha)\ln(1-\alpha)$. Holding factor prices fixed therefore implies $d \ln MC = -d \ln z$. More generally, the same one-for-one relationship holds for Hicks-neutral productivity under constant returns to scale. With non-constant returns

in Table 6, a 1% increase in R&D employment is associated with approximately a 0.019–0.020% increase in productivity and a 0.017–0.018% increase in markups. These estimates imply that a large fraction of the measured productivity gain is associated with higher markups, consistent with incomplete pass-through of innovation-induced efficiency gains into output prices.

To sum up, the evidence presented so far suggests that innovative firms tend to be larger, more productive, and charge higher markups relative to other firms operating in the same industries. Importantly, these relationships remain robust after controlling for observable differences in firm characteristics and measures of operating cost structure, suggesting that they do not simply reflect scale effects or differences in fixed costs. Moreover, the results suggest incomplete pass-through of innovation-driven reductions in marginal costs into output prices and are consistent with environments in which productivity improvements are partially absorbed into firms' markups.

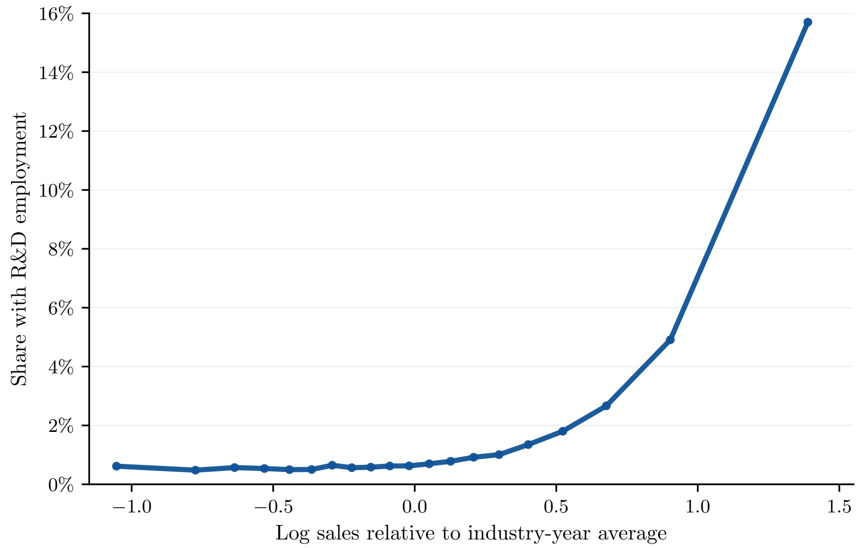
Table 6. R&D Employment, Productivity, and Markups

	$\ln(\text{Productivity}_{it})$		$\ln(\text{Markup}_{it})$	
	(1)	(2)	(3)	(4)
$\ln(\text{R\&D emp}_{it})$	0.0199**	0.0187**	0.0180*	0.0165**
	(0.0096)	(0.0087)	(0.0094)	(0.0081)
Industry FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y
Overhead ratio	-	Y	-	Y
Observations	13,478	13,478	13,478	13,478
R-squared	0.8243	0.8358	0.3842	0.4288

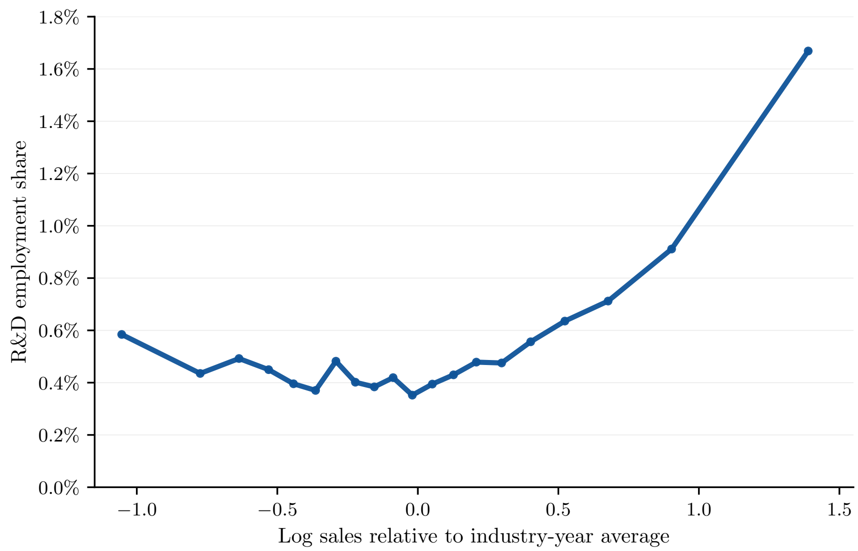
Notes: Columns (1) and (2) use the log of firm-level productivity estimated following De Loecker and Warzynski (2012) under a Cobb–Douglas production function as the dependent variable. Columns (3) and (4) use the log of firm-level markups estimated following De Loecker and Warzynski (2012). All specifications include 2-digit industry and year fixed effects. Firm-level controls include firm age and size (measured by the log of total assets), exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. Columns (2) and (4) additionally control for overhead relative to total expenses as a proxy for fixed operating costs. Table B15 further examines the robustness of the results to replacing separate industry and year fixed effects with industry-by-year fixed effects. Robust standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

to scale, productivity still lowers marginal costs, but the elasticity of marginal cost with respect to productivity need not equal -1.

Figure 2. R&D Employment and Firm Size



(a) Share with R&D employment



(b) R&D employment share

Notes: The figure plots R&D activity by relative firm size. Firm size is measured as log sales relative to the average log sales within the same 2-digit industry-year cell. Firms are first grouped into ventiles based on relative firm size. Within each ventile, Panel (a) reports the share of firms with positive R&D employment, while Panel (b) reports the average firm-level ratio of R&D employment to total employment. The corresponding figures by sector group are reported in Figure A11.

3.2 Innovation and Financial Structure

Next, I investigate whether innovative firms differ systematically from non-innovative firms in their financing structure and explore how firms finance innovation activities. Innovation projects are often difficult to finance externally because their returns are uncertain and frequently tied to intangible assets that are difficult to pledge as collateral. As a result, innovative firms may rely more heavily on internal sources of funding and maintain different balance sheet structures relative to otherwise similar firms. I begin by comparing leverage and equity positions across innovative and non-innovative firms and then examine the relationship between firms' internal funding capacity and innovation activity directly.

To analyze how financing structures differ across firms, I estimate conditional differences in leverage and equity ratios by relating firms' financing positions to an indicator for whether the firm employs R&D workers while controlling for firm size, age, exporter status, legal form, geographic location, and industry and year fixed effects. This approach isolates differences in financing structure associated with innovation activity among firms operating in similar industries and sharing similar observable characteristics. Table 7 reports the results. Innovative firms exhibit systematically lower leverage ratios and higher equity ratios relative to otherwise observably similar firms. In columns (1) and (4), firms employing R&D workers have leverage ratios that are approximately 1.4 percentage points lower and equity ratios that are about 1.3 percentage points higher. Relative to the sample mean leverage ratio of 0.22 (Table 1), the estimates imply that innovative firms rely on debt financing roughly 6.6% less than otherwise comparable non-innovative firms. Similarly, the estimated increase in the equity ratio corresponds to approximately 3.2% of the sample mean equity ratio.

One possible explanation for these patterns is that innovative firms simply hold larger liquidity buffers and therefore have less need for debt financing. To investigate this possibility, columns (2) and (5) additionally control for the cash ratio. The coefficient on the R&D indicator remains statistically significant after accounting for liquidity holdings, mitigating the concern that innovative firms hold less debt because they are cash-rich and have less need for debt.

A second explanation is that innovative firms hold a larger share of intangible assets, which are more difficult to pledge as collateral in external credit markets. I address this possibility by additionally controlling for the collateral ratio in columns (3) and (6). The coefficient on the R&D indicator remains statistically significant after controlling for collateralizability, indicating that innovative firms maintain lower leverage and higher equity ratios for reasons beyond observable differences in tangible asset holdings. Moreover, in a robustness exercise, I decomposes firms' debt positions according to debt maturity structure by separately examining short-term and long-term leverage ratios in Table B17. The results show that the lower leverage ratios of innovative firms reflect reductions in both short-term and long-term debt, although the overall effect is driven primarily by differences in long-term borrowing. In particular, firms employing R&D workers exhibit short-term leverage ratios that are approximately 0.5 percentage points lower, compared with reductions of roughly 1.3 percentage points in long-term leverage. Overall, the results support the interpretation that innovative firms face lower debt capacity.

Table 7. Innovation Activity and Firm Financing Structure

	Leverage ratio_{it}			Equity ratio_{it}		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}(\text{R\&D emp}_{it} > 0)$	-0.0143***	-0.0122***	-0.0084**	0.0135***	0.0100**	0.0084*
	(0.0036)	(0.0035)	(0.0033)	(0.0045)	(0.0047)	(0.0047)
Industry FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y	Y	Y
Cash ratio	-	Y	Y	-	Y	Y
Collateral ratio	-	-	Y	-	-	Y
Observations	1,074,551	1,074,551	1,074,551	1,074,551	1,074,551	1,074,551
R-squared	0.0587	0.1124	0.1505	0.1191	0.2136	0.2184

Notes: Columns (1)–(3) use the leverage ratio, defined as total debt relative to total assets, as the dependent variable. Columns (4)–(6) use the equity ratio, defined as total equity relative to total assets. $\mathbb{1}(\text{R\&D emp} > 0)$ is an indicator equal to one if the firm employs R&D workers. All specifications include 2-digit industry and year fixed effects. Firm-level controls include firm age and size (measured by the log of total assets), exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. The cash ratio is defined as cash and cash equivalents relative to total assets. The collateral ratio is defined as tangible fixed assets relative to total assets. Columns (3) and (6) additionally control for both the cash ratio and the collateral ratio. Table B17 further decomposes the leverage differences into short-term and long-term leverage. Robust standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The previous results suggest that innovative firms rely relatively less on debt financing and maintain stronger equity positions. This pattern naturally raises the question of whether internal funding capacity plays an important role in supporting innovation activity directly. If external financing is comparatively limited for innovative firms, firms with greater internal resources should be more innovative. I now examine this relationship directly by studying how firms' equity positions relate to both the extensive and intensive margins of innovation activity. To do so, I estimate the following regressions:

$$\text{Innovation Activity}_{it} = \beta_0 + \beta_1 \ln(\text{Equity}_{it}) + \Gamma' Z_{it} + \delta_s + \delta_t + \varepsilon_{it}, \tag{6}$$

where Innovation Activity_{it} denotes either an indicator for whether the firm employs R&D workers or the share of R&D employment in total employment, $\ln(\text{Equity}_{it})$ is the log of firm equity, Z_{it} is a vector of firm-level controls, δ_s denotes industry fixed effects, and δ_t denotes year fixed effects.

In additional specifications, I also control for firms' leverage and collateral ratios to isolate the relationship between internal funding capacity and innovation activity conditional on observable differences in external financing capacity. Table 8 reports the corresponding results. In columns (1) and (2), firms with larger equity positions are systematically more likely to employ R&D workers, while columns (3) and (4) show that firms with greater equity also devote a larger share of their workforce to innovation activities. Quantitatively, the estimates imply that a one-log-point increase in firm equity is associated with approximately a 0.19–0.20 percentage point increase in the probability of employing R&D workers and a 0.06–0.07 percentage point increase in the share of workers allocated to R&D activities. Relative to the sample mean R&D participation rate of 1.7%, these estimates correspond to an increase of roughly 11–12%. Similarly, relative to the average R&D employment share of 0.5%, the estimated effect on the intensive margin corresponds to an increase of approximately 12–14%. Moreover, these relationships remain positive and statistically significant after controlling for firms' leverage and collateral ratios, indicating that the relationship between internal funding capacity and innovation activity is not explained by observable differences in external financing positions or collateralizability. The evidence therefore suggests that firms with greater internal funding capacity are more likely to engage in innovation activities and invest more intensively in R&D.

Finally, it is useful to consider whether the lower leverage ratios of innovative firms reflect financing constraints or differences in financing preferences. One possible interpretation is that innovative firms optimally choose to rely less on debt financing because innovation projects are riskier or because they value financial flexibility. Under this view, lower leverage would reflect differences in financing preferences rather than limitations in firms' access to external finance. Several features of the empirical evidence, however, are difficult to reconcile with the debt aversion explanation. First, innovation activity is strongly related to firms' internal funding capacity. If innovative firms could freely substitute toward external financing whenever profitable opportunities arose, one would expect innovation activity to be less sensitive to the availability of internal funds. In that case, equity accumulation would primarily reflect a financing choice rather than a determinant of firms' ability to innovate. Instead, the results show that firms with larger equity positions are systematically more likely to engage in innovation activities and allocate a larger share of their workforce to R&D. This pattern suggests that internally generated funds play an important role in enabling innovation activity rather than merely serving as a preferred financing source. Moreover, innovative firms do not appear to face substantially worse borrowing terms conditional on obtaining external finance. As shown in Table B18, innovative firms exhibit slightly lower interest rates on debt. This pattern suggests that financing frictions associated with innovation operate primarily through quantities rather than prices. Taken together, these findings support the view that innovation depends importantly on internally generated funds and that financing frictions primarily operate through limitations on debt capacity and the quantity of external finance available to innovative firms.

Table 8. Equity and Innovation Activity

	$\mathbb{1}(\text{R\&D emp}_{it} > 0)$		$\frac{\text{R\&D emp}_{it}}{\text{Employment}_{it}}$	
	(1)	(2)	(3)	(4)
$\ln(\text{Equity}_{it})$	0.0020***	0.0019***	0.0007***	0.0006***
	(0.0004)	(0.0005)	(0.0002)	(0.0002)
Industry FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y
Collateral ratio	-	Y	-	Y
Leverage ratio	-	Y	-	Y
Observations	1,074,551	1,074,551	1,062,142	1,062,142
R-squared	0.1228	0.1229	0.0208	0.0209

Notes: The table reports estimates of the relationship between firm equity and innovation activity. The dependent variable in columns (1) and (2) is an indicator equal to one if the firm employs R&D workers. Columns (3) and (4) use the share of R&D employment in total employment as the dependent variable. All specifications include 2-digit industry and year fixed effects. Firm-level controls include firm age and size (measured by the log of total assets), exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. Columns (2) and (4) additionally control for the collateral ratio, defined as the ratio of tangible assets to total fixed assets, and the leverage ratio, defined as total debt relative to total assets. Robust standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

3.3 Market Power and Internal Financing

The evidence presented so far suggests that innovation activities depend importantly on internally generated funds. This raises the question of how firms generate the internal resources needed to finance innovation activities in the presence of financial frictions. Combined with the earlier finding that innovative firms exhibit higher markups on average, one possibility is that market power supports innovation indirectly through its effect on firms' internal financing capacity, allowing firms to accumulate internal funds more rapidly. I first study the relationship between market power and internal financing in the overall sample and then examine whether the same relationship remains operative among innovative firms. To test this mechanism directly, I estimate regressions of the following form:

$$\Delta \ln(\text{Equity}_{it}) = \beta_0 + \beta_1 \ln(\text{Markup}_{it}) + \Gamma' Z_{it} + \delta_s + \delta_t + \varepsilon_{it}, \tag{7}$$

where $\Delta \ln(\text{Equity}_{it})$ denotes the growth rate of firm equity, $\ln(\text{Markup}_{it})$ is the log of firm-level markups estimated from the production function, Z_{it} is a vector of firm-level controls, and δ_s and δ_t denote industry and year fixed effects, respectively. Table 9 reports the results. Columns (1) and (2) present estimates for the full sample, while columns (3) and (4) restrict the sample to innovative firms. Across all specifications, firms with higher markups exhibit systematically faster equity accumulation. For the full sample, the estimated coefficients imply that a 1% increase in firm-level markups is associated with approximately a 0.096 percentage point increase in equity growth. Among innovative firms, a 1% increase in firm-level markups is associated with approximately a 0.076 percentage point increase in equity growth.

Table 9. Market Power and Equity Accumulation

	$\Delta \ln(\text{Equity}_{it})$			
	All firms		*Innovative firms*	
	(1)	(2)	(3)	(4)
$\ln(\text{Markup}_{it})$	0.0955***	0.0920***	0.0757***	0.0682***
	(0.0124)	(0.0123)	(0.0124)	(0.0119)
Industry FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y
Collateral ratio	-	Y	-	Y
Leverage ratio	-	Y	-	Y
Observations	1,550,506	1,550,506	13,478	13,478
R-squared	0.0341	0.0451	0.0399	0.0472

Notes: The dependent variable is the growth rate of firm equity, measured as the first difference of log equity. Firm-level markups are estimated following De Loecker and Warzynski (2012) under a Cobb–Douglas production function specification. Innovative firms are defined as firms employing R&D workers. All specifications include 2-digit industry and year fixed effects. Firm-level controls include firm age and size (measured by the log of total assets), exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. Columns (2) and (4) additionally control for the collateral ratio, defined as the ratio of tangible assets to total fixed assets, and the leverage ratio, defined as total debt relative to total assets. Table B19 further examines the robustness of the results to controlling for investment dynamics, while Table B20 additionally introduces industry-by-year and firm fixed effects. Robust standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Before concluding this section, I discuss several alternative interpretations of the empirical patterns documented above and evaluate their plausibility. A first concern is that high-markup firms accumulate equity faster because their cash flows allow them to access more external finance. This possibility is related to Lian and Ma (2021), who show that corporate borrowing constraints are often based on operating cash flows rather than collateral values. Using detailed data on U.S. corporate debt contracts, they document that most borrowing is governed by cash-flow-based covenants that limit debt relative to earnings. This observation is important for interpreting the relationship between market power and internal finance. If lenders condition borrowing capacity on operating cash flows, firms with higher markups may be able to obtain more external financing because market power generates larger cash flows. In that case, a positive relationship between markups and equity growth could partly reflect differences in debt capacity rather than the direct accumulation of retained earnings. To investigate this possibility, I additionally control for firms' leverage and collateral ratios in columns (2) and (4) of Table 9. The coefficient changes little and remains statistically significant, indicating that high-markup firms accumulate equity faster even conditional on their debt positions. Thus, market power appears to strengthen firms' balance sheets through retained equity, over and above any effect it may have on external borrowing capacity.

I next examine whether higher equity accumulation may simply reflect lower investment rather than stronger internal financing capacity. Under this interpretation, firms with higher markups may accumulate equity mechanically because they retain profits instead of expanding investment or undertaking innovation activities. The broader empirical evidence argues against this explanation. Earlier results from Table 8 show that firms with larger equity positions are systematically more likely to employ R&D workers and devote a larger share of their workforce to innovation activities. Thus, internal funds appear closely linked to firms' innovation behavior rather than merely reflecting passive balance-sheet accumulation. Moreover, as a robustness exercise, Table B19 further addresses this concern by additionally controlling for the investment rate, measured as the first difference of log capital. The relationship between markups and equity accumulation remains positive and statistically significant even after conditioning on investment dynamics, suggesting that the results are not driven by lower investment.

Another concern is that firms with higher markups operate in mature or less competitive industries characterized by more stable profitability and lower financing needs. However, the inclusion of industry fixed effects limits this interpretation. The results do not arise from comparisons across sectors with different competitive environments, but rather from differences across firms operating within the same industry. Thus, the estimates capture the relationship between market power and internal financing conditional on common industry-level conditions. As an additional robustness exercise, columns (1) and (2) of Table B20 includes industry-by-year fixed effects, thereby absorbing common industry shocks and differences in maturity across industries. The positive relationship between markups and equity accumulation remains robust in these specifications, ensuring that the results are not driven by industry trends.

Finally, it is also possible that the estimated relationship between markups and equity accumulation simply reflects underlying differences in firm productivity rather than a financing mechanism. More productive firms may simultaneously exhibit higher markups, generate larger profits, and accumulate equity more rapidly. While this interpretation is difficult to rule out completely, the markup estimates themselves are obtained from production function estimation procedures that net out observed input contributions, which suggest that the results capture more than cross-sectional productivity differences. Moreover, columns (3) and (4) of Table B20 additionally introduce firm fixed effects, thereby exploiting within-firm variation over time. The relationship between markups and equity accumulation remains positive and statistically significant, indicating that the results are not driven by persistent differences in firm productivity across firms.

Overall, the evidence is consistent with the view that market power affects innovation not only through static incentives to innovate, but also through firms' dynamic financing capacity.

4 Model

I develop a quantitative general equilibrium model with heterogeneous firms, endogenous innovation, financial frictions, and variable markups. The model is designed to rationalize the empirical patterns documented in Section 3. Innovative firms are larger, more productive, and charge higher markups. Innovation is also linked to firms' internal financial resources, because firms face borrowing constraints and must finance R&D expenditures before revenues are realized. The key mechanism is the interaction between innovation, market power, and internal finance. Firms that invest in R&D become more productive and expand their market shares. Since demand is non-CES, larger firms face lower demand elasticities and charge higher markups. Higher markups raise profits and allow firms to accumulate internal funds more rapidly. Because internal funds relax borrowing constraints, market power affects innovation not only through current profits, but also through firms' dynamic ability to finance future expansion.

Environment Time is discrete and indexed by $t = 0, 1, \ldots$. The economy is inhabited by a unit mass of infinitely lived firms, indexed by $i = 1, \ldots$, that produce differentiated varieties, and a fixed mass $\bar{L}$ of hand-to-mouth workers who supply labor inelastically.

Preferences Firm owners maximize expected lifetime utility given by

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t u(c_{it}),$$

where β is the discount factor. Period utility is given by $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, which is strictly increasing and concave in consumption c_{it}, with γ representing the coefficient of relative risk aversion.

Technology Firms can operate one of two production technologies: a traditional technology and an R&D-intensive technology. The traditional technology, denoted by τ, refers to production that relies solely on labor and capital. In contrast, R&D-intensive technology, denoted κ, additionally allows firms to allocate labor toward innovation activities.

Traditional technology The production function with traditional technology is a Cobb-Douglas, constant returns-to-scale function

$$y_{it}^{\tau} = A^{\tau}(z_{p,it})\, k_{it}^{\alpha} l_{it}^{1-\alpha}, \tag{8}$$

where y_{it} denotes physical output, $A^{\tau}(z_{p,it}) = \exp(z_{p,it})$ is the firm's productivity, k_{it} is the capital stock, l_{it} is labor, and α is a parameter controlling the elasticity of output to capital. Given factor prices w_t and r_t, profits under the traditional technology are

$$\pi_{it}^{\tau} = p_{it} y_{it}^{\tau} - w_t l_{it} - (r_t + \delta) k_{it}, \tag{9}$$

where p_{it} is the price of its variety, and δ is the rate of depreciation of capital.

R&D technology A firm operating the R&D technology allocates labor both to production and to R&D activities. Output is given by

$$y_{it}^{\kappa} = A^{\kappa}(z_{p,it}, z_{r,i}, v_{it})\, k_{it}^{\alpha} l_{it}^{1-\alpha}, \tag{10}$$

where firm productivity depends on idiosyncratic productivity shocks and endogenous R&D effort

$$A^{\kappa}(z_{p,it}, z_{r,i}, v_{it}) = \exp\left(z_{p,it} + z_{r,i} g(v_{it})\right).$$

The variable $z_{r,i}$ captures persistent heterogeneity in the firm's R&D efficiency. Firms with higher $z_{r,i}$ obtain larger productivity gains from a given amount of R&D labor, so they are more likely to engage in intensive R&D activity. This specification is related to firm-dynamics models in which firms differ in their innovation capacity and the returns to R&D investment (Klette and Kortum, 2004; Acemoglu et al., 2018; Caggese, 2019). The function $g(v_{it})$ governs the productivity contribution of R&D labor. Motivated by Jones (2009) and Bloom et al. (2020), I assume that $g(v_{it})$ is increasing and concave, satisfying $g'(v_{it}) > 0$ and $g''(v_{it}) < 0$. This captures the "burden of knowledge" phenomenon, whereby successive innovation improvements become progressively harder to achieve. Specifically, I adopt the specification

$$g(v_{it}) = \xi \frac{(1 + v_{it})^{1-\nu} - 1}{1 - \nu}$$

where ξ is a scale parameter and ν governs the curvature of the R&D technology. Higher values of ν imply stronger diminishing returns to R&D labor, reducing the marginal effectiveness of innovation effort as firms scale up R&D activities.

The firm's optimal R&D choice equates the marginal productivity gain from additional R&D labor to its marginal cost:

$$z_{r,i} g'(v_{it})\, MR_{it}\, y_{it}^{\kappa} = (1 + \phi) w_t,$$

where MR_{it} denotes marginal revenue. Hence, the optimal level of innovation activity is increasing in firm productivity and R&D ability, but decreasing in the marginal cost of R&D labor.

Taking the path of r_t and w_t as given, profits under the R&D-intensive technology are

$$\pi_{it}^{\kappa} = p_{it} y_{it}^{\kappa} - w_t l_{it} - (1+\phi) w_t v_{it} - (r_t + \delta) k_{it} - \kappa, \tag{11}$$

The parameter $\phi \geq 0$ allows R&D labor to be more costly than production labor, capturing wage premia, specialized skills, or organizational costs associated with innovation activities. The presence of the fixed operating cost $\kappa > 0$ introduces a non-convexity in the firm's innovation decision, rendering the R&D-intensive technology feasible only if operated above a minimum scale. The traditional technology is nested within the R&D-intensive technology as the limiting case $v_{it} = 0$ and $\kappa = 0$.

Market Structure The economy features monopolistic competition in differentiated intermediate goods. Each firm i is the sole supplier of a given variety. There is a perfectly competitive final good firm that produces the homogeneous output good Y_t by assembling all available varieties:

$$\int_0^{N_t} \Upsilon\left(\frac{y_{it}}{Y_t}\right) di = 1$$

where $\Upsilon(\cdot)$ is the Kimball aggregator, which is strictly increasing and concave, that is, $\Upsilon' > 0$, $\Upsilon'' < 0$, with $\Upsilon(1) = 1$. Following Klenow and Willis (2016), I adopt the following specification for the Kimball aggregator:

$$\Upsilon(x) = 1 + (\theta - 1) \exp\left(\frac{1}{\epsilon}\right) \epsilon^{\frac{\theta}{\epsilon} - 1} \left(\Gamma\left(\frac{\theta}{\epsilon}, \frac{1}{\epsilon}\right) - \Gamma\left(\frac{\theta}{\epsilon}, \frac{x^{\frac{\epsilon}{\theta}}}{\epsilon}\right)\right)$$

where $\Gamma(s,z) \equiv \int_z^{\infty} t^{s-1} e^{-t} dt$ is the upper incomplete gamma function. The parameter $\theta > 1$ governs the baseline elasticity of substitution, while $\epsilon \geq 0$ controls the degree of demand curvature. In the limit as $\epsilon \to 0$, the Kimball aggregator converges to the standard CES case.

The final good producer maximizes profits by choosing quantities y_{it}, taking prices p_{it} as given:

$$\max_{\{y_{it}\}} Y_t - \int_0^{N_t} p_{it} y_{it} di$$

subject to the Kimball aggregator (4). The solution to this problem gives rise to the inverse demand function:

$$\frac{p_{it}}{P_t} = D_t \Upsilon'\left(\frac{y_{it}}{Y_t}\right) = D_t \left(\frac{\theta - 1}{\theta}\right) \exp\left(\frac{1 - \left(\frac{y_{it}}{Y_t}\right)^{\epsilon/\theta}}{\epsilon}\right),$$

where P_t is the aggregate price index and D_t is an endogenous demand index. The elasticity of demand faced by the firm is:

$$\sigma(x) = -\frac{\Upsilon'(x)}{\Upsilon''(x)x} = \theta x^{-\epsilon/\theta},$$

while the superelasticity of demand is

$$-\frac{d\ln\sigma(x)}{d\ln x} = \frac{\epsilon}{\theta}.$$

Firms therefore set markups according to

$$\mu(x) = \frac{\sigma(x)}{\sigma(x)-1} = \frac{\theta}{\theta - \left(\frac{y_{it}}{Y_t}\right)^{\epsilon/\theta}}.$$

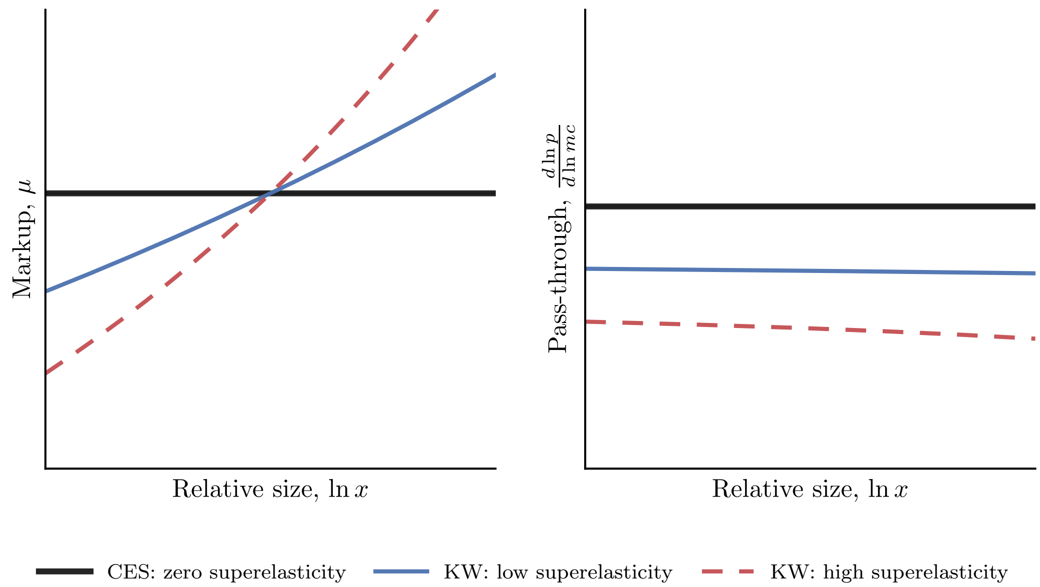
Unlike the CES benchmark, demand elasticity varies endogenously with firm size. Larger firms face lower demand elasticities and therefore set higher markups. An additional implication of Kimball demand concerns incomplete pass-through of marginal costs into prices. Since markups vary with firm size, marginal cost changes induce endogenous markup adjustments. Log-differentiating the pricing condition, $p(x) = \mu(x)mc$, gives:

$$\rho(x) \equiv \frac{d\ln p(x)}{d\ln mc} = \frac{1}{1+\sigma(x)\frac{d\ln\mu(x)}{d\ln x}} = \frac{1}{1+\frac{\epsilon}{\theta}\frac{\sigma(x)}{\sigma(x)-1}}.$$

which is strictly below one whenever $\epsilon > 0$.

Figure 3 illustrates the consequences of Kimball demand for firm-level markups and pass-through. The left panel shows markups increase with firms' relative size. Since larger firms face lower demand elasticities, they optimally charge higher markups. The strength of this relationship is governed by the superelasticity parameter. When ϵ/θ is high, markups rise more steeply with market share, generating substantial heterogeneity in market power across firms. By contrast, the CES benchmark implies constant markups that are independent of firm size. The right panel shows the implications for pass-through. Under CES demand, productivity improvements are fully passed through into lower prices because markups remain constant. Under Kimball demand, however, productivity improvements increase firms' market shares and therefore raise markups endogenously. As a result, firms only partially pass marginal cost reductions through to prices. The degree of incomplete pass-through is increasing in the superelasticity parameter ϵ/θ. When superelasticity is high, firms absorb a larger fraction of productivity gains through higher markups, implying lower pass-through elasticities. In the limit as $\epsilon \to 0$, the model converges to the CES benchmark with constant elasticities, constant markups, and complete pass-through.

Figure 3. Variable Markups and Incomplete Pass-Through under Kimball Demand



Productivity Firms are subject to three sources of productivity heterogeneity: a transitory productivity component, a persistent R&D ability component, and a stochastic innovation-opportunity component. There is no aggregate uncertainty.

The transitory firm productivity evolves according to an AR(1) process,

$$z_{p,it} = \rho z_{p,it-1} + \sigma \varepsilon_{it}, \qquad \varepsilon_{it} \sim N(0,1),$$

where ρ measures the degree of persistence in productivity and σ governs the variance of idiosyncratic productivity risk.

Firms also differ in their persistent R&D ability $z_{r,it}$, which determines how efficiently R&D labor is transformed into productivity gains. R&D ability is drawn from a Pareto distribution,

$$z_{r,it} \sim \text{Pareto}(b_r),$$

where $b_r > 0$ governs the thickness of the upper tail of the R&D ability distribution. Lower values of b_r imply greater dispersion in innovation ability and a higher probability of extremely innovative firms.

Financial Markets Financial markets are incomplete. Firms can save and borrow in a one-period bond b_{it} that pays a real interest rate r_t, but borrowing is limited by internal net worth. As a result, firms can only borrow intertemporally up to a fraction of their capital stock. The borrowing constraint is given by

$$b_{it} \leq \chi k_{it},$$

where b_{it} denotes debt and $\chi \geq 0$ indexes the tightness of the borrowing constraint. A higher value of χ relaxes borrowing limits by allowing firms to pledge a larger fraction of capital as collateral. When $\chi = 0$, firms operate in a zero-credit environment and must rely entirely on internal funds. As $\chi \to \infty$, borrowing constraints cease to bind and firms become financially unconstrained.

Letting $a_{it} = k_{it} - b_{it}$ denote the firm's equity, the borrowing constraint can equivalently be written as

$$k_{it} \leq \lambda a_{it}, \tag{12}$$

where $\lambda \equiv \frac{1}{1-\chi}$. Hence, tighter financial frictions reduce the scale at which firms can operate for a given level of equity.

Recursive Representation Let the individual state vector be $\mathbf{x} = (a, z_p, z_r)$, and let the aggregate state vector be $\mathbf{\Omega} = (w, r, P, Y, D)$. Firms take $\mathbf{\Omega}$ as given. Using primes to denote next-period variables, the firm's recursive problem can be written as

$$V(\mathbf{x}; \mathbf{\Omega}) = \max \left\{ V^{\tau}(\mathbf{x}; \mathbf{\Omega}), V^{\kappa}(\mathbf{x}; \mathbf{\Omega}) \right\}.$$

Under the traditional technology, the firm solves

$$V^{\tau}(\mathbf{x}; \mathbf{\Omega}) = \max_{a'} \left\{ u(c) + \beta \mathbb{E} \left[V(\mathbf{x}'; \mathbf{\Omega}) \mid z_p, z_r \right] \right\}, \tag{13}$$

subject to

$$c = (1+r)a + \pi^{\tau}(\mathbf{x}; \mathbf{\Omega}) - a', \tag{14}$$

and $c \geq 0$. Static profits $\pi^{\tau}(\mathbf{x}; \mathbf{\Omega})$ are given by (9), subject to (4), (8), (12), $k \geq 0$, and $l \geq 0$.

Under the R&D-intensive technology, the firm solves

$$V^{\kappa}(\mathbf{x}; \mathbf{\Omega}) = \max_{a'} \left\{ u(c) + \beta \mathbb{E} \left[V(\mathbf{x}'; \mathbf{\Omega}) \mid z_p, z_r \right] \right\}, \tag{15}$$

subject to

$$c = (1+r)a + \pi^{\kappa}(\mathbf{x}; \mathbf{\Omega}) - a', \tag{16}$$

and $c \geq 0$. Static profits $\pi^{\kappa}(\mathbf{x}; \mathbf{\Omega})$ are given by (11), subject to (4), (10), (12), $k \geq 0$, $l \geq 0$, and $v \geq 0$.

Decision rules Firms face two types of decisions: static input choices and dynamic asset accumulation. The static problem determines the firm's optimal scale, market share, markup, and innovation intensity for a given level of equity and productivity. The dynamic problem determines how much of current resources the entrepreneur consumes and how much is retained inside the firm.

Entrepreneurs are financially constrained and therefore accumulate assets to relax future borrowing limits and expand production when favorable productivity shocks arrive. As a result, both current profits and the prospect of future profits affect firms' incentives to retain earnings inside the firm. This dynamic saving decision is central to the model because innovation raises productivity and market shares, while larger market shares generate higher markups. Higher markups increase profits and accelerate the accumulation of internal funds, thereby strengthening firms' ability to finance future expansion and innovation activities. The economy's competitive structure therefore plays a central role in shaping firms' dynamic incentives through its effect on market shares, markups, and internal financing capacity.

The firm takes aggregate variables as given, but recognizes that its output choice affects the price of its own variety through the demand curve. Letting $q \equiv y/Y$ denote the firm's relative market share, marginal revenue is given by

$$MR(q) = p(q)\left(1 - \frac{1}{\sigma(q)}\right) = \frac{p(q)}{\mu(q)}.$$

For the traditional technology, the interior optimality conditions satisfy $MR(q) \cdot MPK^{\tau} = r + \delta$ and $MR(q) \cdot MPL^{\tau} = w$, where $MPK^{\tau} = \alpha y^{\tau}/k$, and $MPL^{\tau} = (1-\alpha)y^{\tau}/l$.

For the R&D-intensive technology, the interior optimality conditions satisfy $MR(q) \cdot MPK^{\kappa} = r + \delta$, $MR(q) \cdot MPL^{\kappa} = w$, and $MR(q) \cdot MPV^{\kappa} = (1+\phi)w$, where $MPK^{\kappa} = \alpha y^{\kappa}/k$, $MPL^{\kappa} = (1-\alpha)y^{\kappa}/l$, and $MPV^{\kappa} = z_r g'(v) y^{\kappa}$.

When the borrowing constraint binds, capital is instead pinned down by (12), and the firm operates at a smaller scale than it would choose absent financial frictions.

The model does not generally admit closed-form decision rules because firms face a non-linear demand system and borrowing constraints. Under Kimball demand, marginal revenue depends on the firm's relative market share $q = y/Y$. As a result, optimal input choices depend not only on factor prices and productivity, but also on the firm's position in the demand system.

For a given level of equity a, the borrowing constraint limits the firm's feasible capital choice. When the constraint binds, capital is given by (12). Conditional on capital, the firm chooses labor and, under the R&D-intensive technology, R&D labor to satisfy the static optimality conditions above. These choices determine output, market share, markup, and current profits.

The dynamic decision rule $a'(\mathbf{x};\mathbf{\Omega})$ determines how much of current resources the entrepreneur retains inside the firm. This decision reflects the value of internal funds. Firms with high productivity or high R&D efficiency have stronger incentives to save because additional internal assets relax future borrowing constraints and allow them to operate at larger scale. Under Kimball demand, larger scale also raises markups by increasing the firm's market share, which further increases the return to internal funds.

The technology-choice rule $\mathcal{T}(\mathbf{x};\mathbf{\Omega}) \in \{\tau,\kappa\}$ compares the value of operating the traditional technology with the value of operating the R&D-intensive technology. Firms choose the R&D technology when the expected gains from higher productivity and future profitability outweigh the additional R&D labor cost and the fixed operating cost. The R&D technology is therefore more attractive for firms with high baseline productivity, high R&D efficiency, and sufficient internal assets to finance the scale required to benefit from innovation.

Equilibrium Let $\mathbf{x} = (a, z_p, z_r)$ denote the individual state vector and let $\mathbf{\Omega} = (w, r, P, Y, D)$ denote the aggregate state vector. A stationary equilibrium consists of an aggregate state vector $\mathbf{\Omega}$, policy functions $(\mathcal{T}(\mathbf{x};\mathbf{\Omega}), k(\mathbf{x};\mathbf{\Omega}), l(\mathbf{x};\mathbf{\Omega}), v(\mathbf{x};\mathbf{\Omega}), p(\mathbf{x};\mathbf{\Omega}), c(\mathbf{x};\mathbf{\Omega}), a'(\mathbf{x};\mathbf{\Omega}))$, value functions $(V(\mathbf{x};\mathbf{\Omega}), V^{\tau}(\mathbf{x};\mathbf{\Omega}), V^{\kappa}(\mathbf{x};\mathbf{\Omega}))$, and an invariant distribution $\mathbf{\Lambda}(\mathbf{x})$, such that:

1. Given $\mathbf{\Omega}$, firms solve the recursive problem above and choose production technology $\mathcal{T}(\mathbf{x};\mathbf{\Omega}) \in \{\tau,\kappa\}$, capital $k(\mathbf{x};\mathbf{\Omega})$, production labor $l(\mathbf{x};\mathbf{\Omega})$, R&D labor $v(\mathbf{x};\mathbf{\Omega})$, variety price $p(\mathbf{x};\mathbf{\Omega})$, consumption $c(\mathbf{x};\mathbf{\Omega})$, and next-period equity $a'(\mathbf{x};\mathbf{\Omega})$.

2. The labor market clears:
$$\int [l(\mathbf{x};\mathbf{\Omega}) + v(\mathbf{x};\mathbf{\Omega})]\, d\mathbf{\Lambda}(\mathbf{x}) = \bar{L}.$$

3. The capital market clears:
$$\int a\, d\mathbf{\Lambda}(\mathbf{x}) = \int k(\mathbf{x};\mathbf{\Omega})\, d\mathbf{\Lambda}(\mathbf{x}).$$

4. Aggregate output satisfies the Kimball aggregation condition:
$$\int \Upsilon\left(\frac{y(\mathbf{x};\mathbf{\Omega})}{Y}\right) d\mathbf{\Lambda}(\mathbf{x}) = 1.$$

5. The demand index satisfies
$$D = \left[\int \frac{y(\mathbf{x};\mathbf{\Omega})}{Y}\Upsilon'\left(\frac{y(\mathbf{x};\mathbf{\Omega})}{Y}\right) d\mathbf{\Lambda}(\mathbf{x})\right]^{-1}.$$

6. The distribution $\mathbf{\Lambda}(\mathbf{x})$ is invariant under the optimal asset accumulation policy $a'(\mathbf{x};\mathbf{\Omega})$ and the exogenous productivity transition processes.

5 Quantitative Analysis

This section brings the model to the data and uses the calibrated economy to quantify the mechanisms developed above. I first describe the calibration strategy and discuss the quantitative fit of my framework with respect to targeted moments obtained from the data. I then validate the calibrated model by examining its performance in matching key moments from the data that were not targeted during the calibration. Finally, I use the calibrated model to quantify the financing role of markup rents and examine how its importance varies with the severity of borrowing constraints.

5.1 Calibration

To parameterize the model, I begin by partitioning the parameter space into two groups. The first group contains parameters fixed externally before the calibration procedure. The second group contains parameters chosen internally to match moments of the empirical firm distribution, innovation activity, financing behavior, and market power.

Table 10 reports the externally fixed parameters. A period in the model corresponds to a year, the same frequency as the firm-level data. I choose a capital share of $\alpha = 0.33$ and a coefficient of relative risk aversion of $\gamma = 1.5$, both standard values in the macroeconomic literature. The depreciation rate is $\delta = 0.06$, as in Buera and Shin (2013). Following Cagetti and De Nardi (2006) and Gopinath et al. (2017), I use a discount factor of $\beta = 0.865$. The aggregate price level P and the aggregate labor supply $\bar{L}$ are normalized to unity.

Table 10. Externally Fixed Parameters

Fixed	Value	Description
α	0.33	Capital share
γ	1.5	Risk aversion
δ	0.06	Depreciation rate
β	0.865	Discount factor
$\bar{L}$	1.00	Labor supply

The remaining parameters are calibrated to match relevant moments in the firm-level data, as reported in Table 11. Although all model parameters simultaneously impact all target moments, below I provide heuristics for mapping the parameters to the moments.

The first group of parameters governs innovation. The scale parameter of R&D technology, $\xi = 0.317$, is disciplined by the relative employment scale of innovative firms. A higher value of ξ increases the productivity gains from innovation, allowing innovative firms to expand relative to non-innovative firms. The calibrated model generates innovative firms that employ 9.2 times more workers than the average firm, compared with 9.8 times in the data. The curvature parameter of R&D technology, $\nu = 4.346$, governs how R&D employment scales with firm size and is targeted to the elasticity of R&D employment with respect to output. Larger values of ν imply stronger diminishing returns to innovation effort and therefore weaken the propensity of large firms to devote more labor to R&D. The calibrated model matches the empirical elasticity, generating a value of 0.36, as in the data.

Table 11. Internally Fitted Parameters

Fitted	Value	Description	Targeted moment	Model	Data
Panel A: Innovation					
ξ	0.317	Scale of R&D techn.	Rel. emp.: innov./all	9.146	9.790
ν	4.364	Curvature of R&D tech.	Elast. $\ln(RD)$ on $\ln(Y)$	0.362	0.359
κ	0.484	Innovation fixed cost	Share innovating firms	0.009	0.020
ϕ	0.238	Innovation labor overhead	R&D emp. share	0.003	0.005
b_r	19.661	Pareto tail parameter	Top 1% R&D emp. share	0.999	0.910
Panel B: Productivity					
ρ	0.901	Persistence of productivity	Top 10% emp. share	0.474	0.660
σ	0.320	Std. dev. prod. shocks	Std. dev. $\ln(Z)$	0.868	0.940
Panel C: Market Structure					
θ	3.467	Baseline elast. of demand	Median markup	1.307	1.370
ε/θ	0.148	Superelasticity of demand	Mean markup	1.315	1.730
Panel D: Financing					
λ	1.683	Borrowing constraint	Mean leverage ratio	0.209	0.220

Notes: This table reports the parameters jointly calibrated inside the model and the targeted moments used in the calibration. Y denotes output, Z denotes firm productivity, and RD denotes R&D employment. Innovative firms are defined as firms with positive R&D employment. In the data, the elasticity of R&D employment with respect to output is estimated from a firm-level log-log regression that includes 2-digit industry and year fixed effects, as well as controls for age, exporter status, legal form, district, single-owner status, and multi-establishment status.

As in Buera et al. (2011), fixed costs introduce non-convexities in the technology adoption decision. The fixed cost of innovation, $\kappa = 0.484$, primarily disciplines the extensive margin of innovation. A higher value of κ raises the scale required for R&D to be profitable, discouraging smaller firms from adopting the R&D-intensive technology and reducing the share of firms that innovate. The labor-overhead parameter, $\phi = 0.238$, instead disciplines the intensive margin by increasing the marginal cost of R&D labor. A higher value of ϕ reduces the amount of labor allocated to innovation among firms that choose to conduct R&D, helping the model match the aggregate R&D employment share. The calibrated model generates an innovation participation rate of 0.9%, compared with 2.0% in the data, and an aggregate R&D employment share of 0.3%, compared with 0.5% in the data.

The Pareto tail parameter governing R&D ability, $b_r = 19.661$, determines the dispersion of innovation opportunities across firms. Smaller values imply a thicker upper tail of R&D ability and therefore greater concentration of innovation activity. I choose b_r to match the concentration of R&D employment among the largest innovators, measured by the share of aggregate R&D employment accounted for by the top one percent of firms. The calibrated model generates a top-1% R&D employment share of 99.9%, compared with 91.0% in the data.

The second group of parameters governs the productivity process. The persistence parameter, $\rho = 0.901$, determines the extent to which productivity differences persist over time and is chosen to match the concentration of employment among the largest firms. Higher values of ρ increase the persistence of productivity shocks, raising concentration in the firm-size distribution. The calibrated model generates a top-10% employment share of 47.4%, compared with 66.0% in the data. The volatility parameter, $\sigma = 0.320$, controls the dispersion of productivity shocks and is disciplined by the cross-sectional dispersion of measured productivity. Larger values of σ generate greater heterogeneity in firm performance and widen the productivity distribution. The calibrated model produces a standard deviation of log productivity equal to 0.87, compared with 0.94 in the data.

The third group of parameters governs market structure. Under variable markups, the markup distribution depends both on the baseline elasticity of demand and on the curvature of demand. The baseline elasticity parameter, θ, is calibrated to match the median markup in the data, while the superelasticity parameter, ε/θ, is disciplined by the mean markup. The calibrated value $\theta = 3.467$ generates a median markup of 1.31, compared with 1.37 in the data. This calibration is intentionally conservative, since higher markups would mechanically strengthen the financing role of market power by allowing firms to accumulate internal funds more rapidly. For comparison, Acemoglu et al. (2018) assume an elasticity of substitution of 2.9, which corresponds to a constant-markup benchmark of approximately 1.52. The superelasticity parameter is calibrated to $\varepsilon/\theta = 0.148$, which generates a mean markup of 1.32, compared with 1.73 in the data. A positive superelasticity implies that demand becomes less elastic as firms expand, allowing larger and more productive firms to charge higher markups. The estimated value is close to those used in recent quantitative models with variable markups. For example, Liang (2023), Boar and Midrigan (2025), and Edmond et al. (2023) calibrate superelasticity parameters of 0.14, 0.15, and 0.16, respectively. Thus, the markup heterogeneity generated by the model is consistent with the literature.

The final parameter governs access to financing. The borrowing constraint parameter, $\lambda = 1.683$, determines the maximum amount of capital that firms can operate for a given level of equity and is calibrated to match the average leverage ratio in the data. The calibrated value lies within the range of borrowing multipliers used in the literature on financial frictions. For example, Moll (2014) considers values ranging from $\lambda = 1.06$ for China to $\lambda = 4.15$ for the United States, reflecting cross-country differences in financial development. The calibrated model generates an average leverage ratio of 0.21, compared with 0.22 in the data.

5.2 Validation

I next evaluate the model using moments that were not directly targeted in the calibration. Table 12 reports these validation moments. Overall, the model successfully matches key dimensions of firm heterogeneity, including the concentration of output among large firms, the aggregate importance of innovative firms, and the size gradient in innovation activity.

Table 12. Untargeted Moments

Moment	Data	Model
Panel A: Concentration moments		
Top 20% output share	0.810	0.718
Top 10% output share	0.680	0.523
Top 5% output share	0.540	0.327
Panel B: Aggregate importance of innovative firms		
Employment share	0.070	0.081
Output share	0.050	0.097
Panel C: Innovation across the size distribution		
Share innovating: top 50%	0.030	0.018
Share innovating: top 20%	0.070	0.044
Share innovating: top 10%	0.110	0.088
Share innovating: top 5%	0.180	0.182
R&D intensity: top 50%	0.007	0.003
R&D intensity: top 20%	0.010	0.004
R&D intensity: top 10%	0.014	0.006
R&D intensity: top 5%	0.017	0.010

Notes: This table reports untargeted moments used to evaluate the quantitative performance of the calibrated model. Output is measured as sales. Innovative firms are defined as firms with positive R&D employment. Concentration moments report the share of aggregate output accounted for by firms in the top percentiles of the output distribution. Panel B reports the share of aggregate employment and output accounted for by innovative firms. Panel C reports innovation participation and R&D intensity within the top percentiles of the employment distribution. R&D intensity is measured as R&D employment divided by total employment.

Panel A reports validation moments for the concentration of economic activity. The model reproduces the concentration of output in the upper tail of the firm-size distribution: the top 20%, 10%, and 5% of firms account for 71.8%, 52.3%, and 32.7% of output, respectively, compared with 81.0%, 68.0%, and 54.0% in the data. Thus, the model captures the strong skewness of output toward large firms.

Turning to Panel B, the model performs well in capturing the aggregate importance of innovative firms. Although innovative firms represent a small fraction of all firms, they account for a disproportionate share of economic activity. In the data, innovative firms account for 7.0% of aggregate employment and 5.0% of output, compared with 8.1% and 9.7% in the model. Thus, the model reproduces the fact that innovation is concentrated among economically important firms that account for a sizable share of aggregate production.

Panel C shows that the model captures the sorting of innovation toward larger firms. In both the model and the data, innovation becomes increasingly prevalent among firms in the upper percentiles of the employment distribution. The share of firms engaged in innovation rises from 1.8% among the top half of firms to 18.2% among the top 5% in the model, tracking the corresponding increase from 3.0% to 18.0% in the data. R&D intensity also rises monotonically with firm size: in the model, it increases from 0.3% among the top half of firms to 1.0% among the top 5%, compared with an increase from 0.7% to 1.7% in the data.

5.3 Decomposing the Financing Role of Markup Rents

This section uses the calibrated model to decompose and quantify the channels through which market power affects innovation and aggregate outcomes. In the baseline economy, markups raise current profits, increase retained earnings, and relax future borrowing constraints. However, market power also affects firms through other margins, including technology choices, factor inputs, and general-equilibrium effects. To isolate the financing channel of market power, I introduce a wedge in the firm's equity accumulation decision that varies the extent to which markup rents can be transformed into equity, while leaving the remaining firm decisions unaffected. I define markup rents as the portion of revenue associated with the gap between price and marginal revenue:

$$\mathcal{M}_{it} = p_{it}y_{it} - MR_{it}y_{it} = \left(1 - \frac{1}{\mu_{it}}\right)p_{it}y_{it},$$

where $MR_{it} = p_{it}/\mu_{it}$. A firm receives p_{it} per unit sold, but because it faces a downward-sloping demand curve, the marginal value of expanding sales is $MR_{it} < p_{it}$. The difference between actual revenue and revenue evaluated at marginal revenue therefore captures the inframarginal revenue generated by market power.

I discipline this channel through an internal-finance wedge $\tau \in [0, 1]$ that controls the share of markup rents that can be carried forward as equity. At the equity accumulation stage, the profits available to be retained as firm equity are given by

$$\tilde{\pi}_{it}(\tau) = \pi_{it} - \tau\mathcal{M}_{it},$$

so that the firm's dynamic budget constraint becomes

$$c_{it} + a_{it+1} = (1 + r_t)a_{it} + \tilde{\pi}_{it}(\tau).$$

The wedge therefore applies only to the resources that can be used as internal funds. When $\tau = 0$, all markup rents can be retained and the economy coincides with the baseline. When $\tau = 1$, markup rents are fully excluded from retained earnings and cannot be used to accumulate equity. Thus, increasing τ restricts the ability of market power to generate internal funds.

To formalize the decomposition, it is useful to distinguish between firms' operating and dynamic decisions. Let

$$\mathbf{g}(\mathbf{x}; \mathbf{\Omega}) = (\mathcal{T}(\mathbf{x}; \mathbf{\Omega}), k(\mathbf{x}; \mathbf{\Omega}), l(\mathbf{x}; \mathbf{\Omega}), v(\mathbf{x}; \mathbf{\Omega}), p(\mathbf{x}; \mathbf{\Omega}), y(\mathbf{x}; \mathbf{\Omega}))$$

denote the vector of operating policies, and let

$$\mathbf{h}(\mathbf{x}; \mathbf{\Omega}) = \big(c(\mathbf{x}; \mathbf{\Omega}), a'(\mathbf{x}; \mathbf{\Omega})\big)$$

denote the vector of dynamic policies. The baseline economy is characterized by $(\mathbf{g}^0, \mathbf{h}^0, \mathbf{\Omega}^0)$. For a given value of the internal-finance wedge τ, let $\mathbf{h}^F(\mathbf{x}; \tau, \mathbf{\Omega})$ denote the optimal dynamic-policy vector when firms continue to use the baseline operating-policy vector $\mathbf{g}^0$. Likewise, $(\mathbf{g}^I(\mathbf{x}; \tau, \mathbf{\Omega}), \mathbf{h}^I(\mathbf{x}; \tau, \mathbf{\Omega}))$ denote the optimal operating and dynamic policies when firms fully internalize the wedge. In the general-equilibrium exercises, aggregate conditions are determined endogenously for each value of τ, yielding $\mathbf{\Omega}(\tau)$, while in the partial-equilibrium exercises aggregate conditions remain fixed at their baseline values $\mathbf{\Omega}^0$.

Table 13. Counterfactual Exercises for Decomposing the Financing Role of Markup Rents

Counterfactual	Operating policies	Dynamic policies	Aggregate state
Financing, PE	$\mathbf{g}^0$	$\mathbf{h}^F(\tau)$	$\mathbf{\Omega}^0$
Financing, GE	$\mathbf{g}^0$	$\mathbf{h}^F(\tau)$	$\mathbf{\Omega}(\tau)$
Financing+Incentives, PE	$\mathbf{g}^I(\tau)$	$\mathbf{h}^I(\tau)$	$\mathbf{\Omega}^0$
Financing+Incentives, GE	$\mathbf{g}^I(\tau)$	$\mathbf{h}^I(\tau)$	$\mathbf{\Omega}(\tau)$

Notes: PE denotes partial equilibrium and GE denotes general equilibrium. The vector $\mathbf{g}$ collects operating policies, while $\mathbf{h}$ collects dynamic policies.

I implement the decomposition along two dimensions, yielding the four counterfactuals summarized in Table 13. The first dimension varies the margin of firm behavior exposed to the wedge. In the financing-only exercises, firms continue to operate according to the baseline policy vector $\mathbf{g}^0$. The wedge enters only through the dynamic problem, so firms re-optimize $\mathbf{h}^F(\mathbf{x}; \tau, \mathbf{\Omega})$, taking into account its effect on retained earnings, future equity, and continuation values. Technology choice, factor demands, prices, output, and R&D effort therefore remain governed by the baseline operating policies. Markup rents continue to influence production and pricing decisions exactly as in the baseline economy, but only a fraction of those rents can be transformed into internal funds, thereby isolating the financing channel of markup rents. In the financing-and-incentives exercises, firms instead solve for $(\mathbf{g}^I(\mathbf{x};\tau,\mathbf{\Omega}), \mathbf{h}^I(\mathbf{x};\tau,\mathbf{\Omega}))$, internalizing the wedge when making both operating and dynamic decisions. These exercises therefore capture both the financing role of markup rents and the effect of a lower return to market power on production, pricing, and innovation incentives. The second dimension concerns the treatment of aggregate conditions. In the partial-equilibrium exercises, firms optimize taking the baseline aggregate state $\mathbf{\Omega}^0$ as given. In the general-equilibrium exercises, the economy is re-solved for each value of τ, yielding an equilibrium aggregate state $\mathbf{\Omega}(\tau)$.[5]

I now use these counterfactuals to quantify the importance of the financing channel for aggregate innovation and output. Figure 4 shows that restricting firms' ability to retain markup rents reduces both outcomes. The key result is that the internal-finance wedge reduces innovation and output even after shutting down changes in firms' production, pricing, and innovation incentives. At $\tau = 0.1$, R&D participation falls by 9.0% in partial equilibrium and 9.4% in general equilibrium, while output falls by about 3.3% in both cases. The effects become larger once firms are allowed to adjust their operating decisions in response to the wedge. At $\tau = 0.1$, R&D participation falls by 78.6% in partial equilibrium and 73.1% in general equilibrium, while output falls by 10.8% and 4.5%, respectively. These additional declines reflect the incentive channel: when firms recognize that a smaller fraction of markup rents can ultimately be retained, the return to expanding market share and generating future rents is reduced, weakening incentives to innovate and produce.

Table 14. Contribution of the Financing Channel for Aggregate Innovation and Output

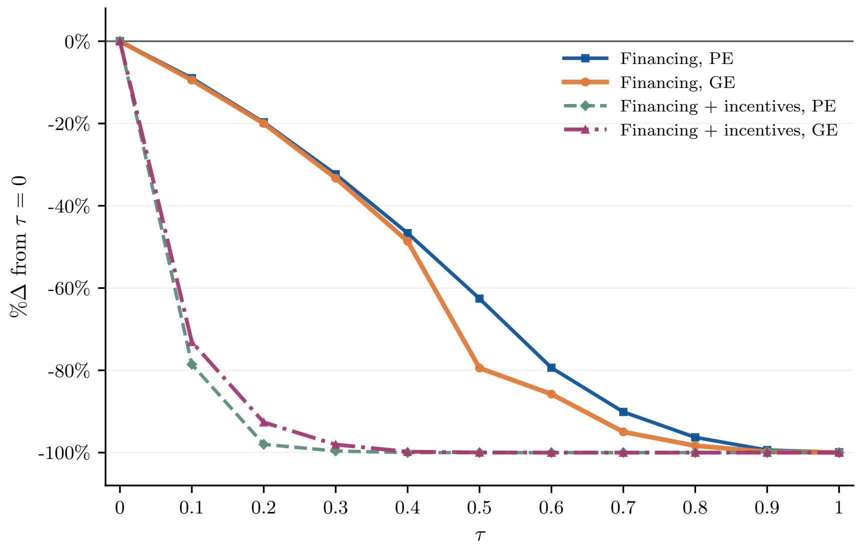
	R&D participation		Output	
τ	PE	GE	PE	GE
0.1	0.11	0.13	0.31	0.74
0.2	0.20	0.22	0.36	0.83
0.3	0.33	0.34	0.42	0.89
0.4	0.47	0.49	0.47	0.93
0.5	0.63	0.80	0.52	0.98

Notes: Entries report the share of the total effect attributable to the financing channel. For each outcome, the share is computed as the ratio of the absolute percentage change in the financing-only exercise to the absolute percentage change in the corresponding financing-and-incentives exercise. PE denotes partial equilibrium and GE denotes general equilibrium.

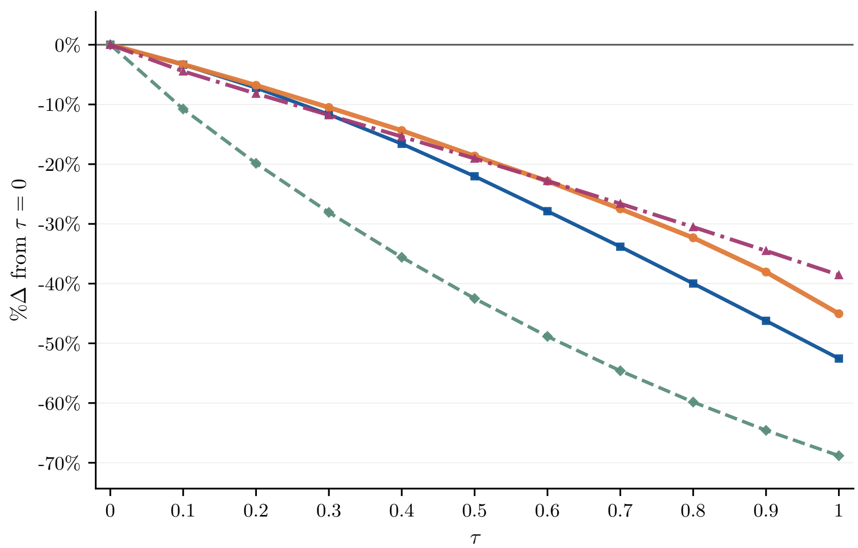
[5] In the financing-only general-equilibrium exercise, firms continue to use the baseline operating-policy vector $\mathbf{g}^0$, so the wedge enters directly only through $\mathbf{h}^F(\mathbf{x};\tau,\mathbf{\Omega})$. Nevertheless, changes in equilibrium wages, interest rates, and demand conditions imply that realized production outcomes can still differ from the baseline through general-equilibrium feedback effects.

Figure 4. Impact on Aggregate Innovation and Output

(a) R&D participation



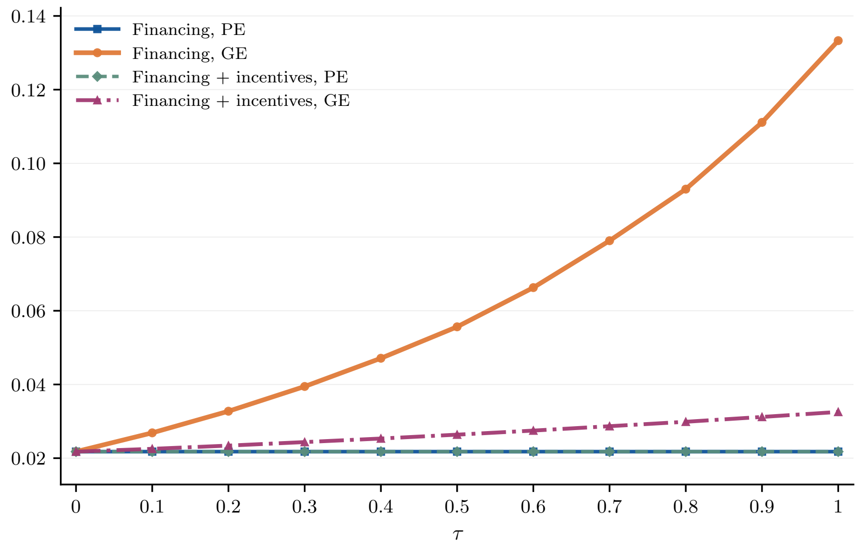
(b) Output



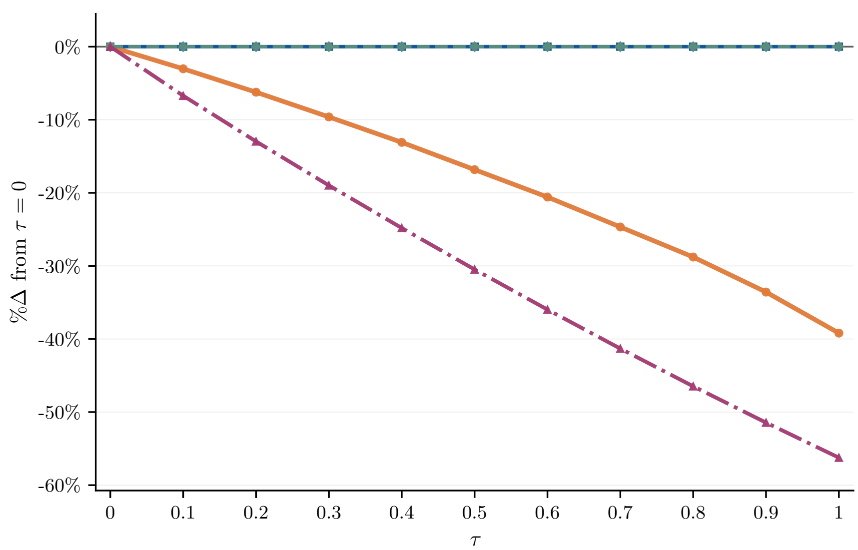
Notes: This figure reports the effects of the internal-finance wedge on aggregate innovation and output. Panel (a) reports the percentage change in the share of R&D-active firms relative to the baseline economy, while Panel (b) reports the percentage change in aggregate output. The response of aggregate R&D labor is similar to the participation margin and is shown in Figure C12.

Figure 5. Impact on Equilibrium Prices

(a) Interest rate



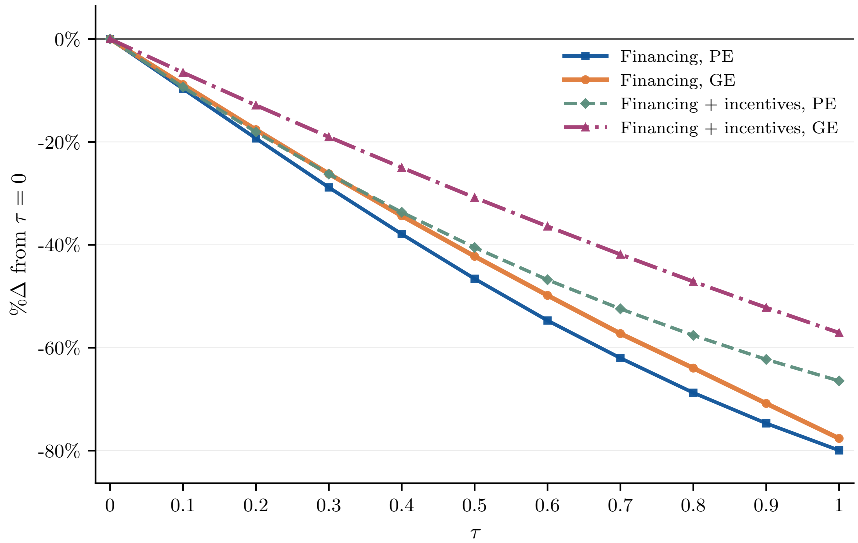
(b) Wage



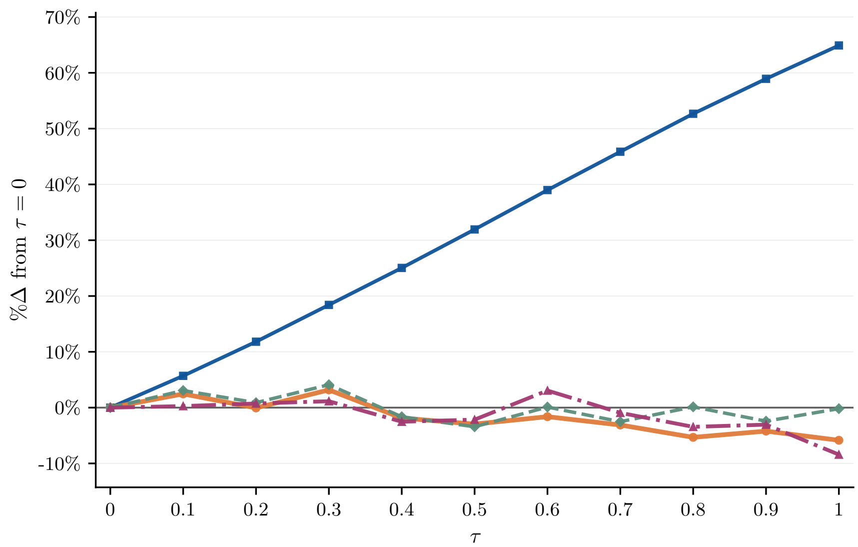
Notes: This figure reports the effects of the internal-finance wedge on equilibrium prices. Panel (a) reports the equilibrium interest rate in levels, while Panel (b) reports the percentage change in the equilibrium wage relative to the baseline economy.

Figure 6. Impact on Equity and Financing Constraints

(a) Equity



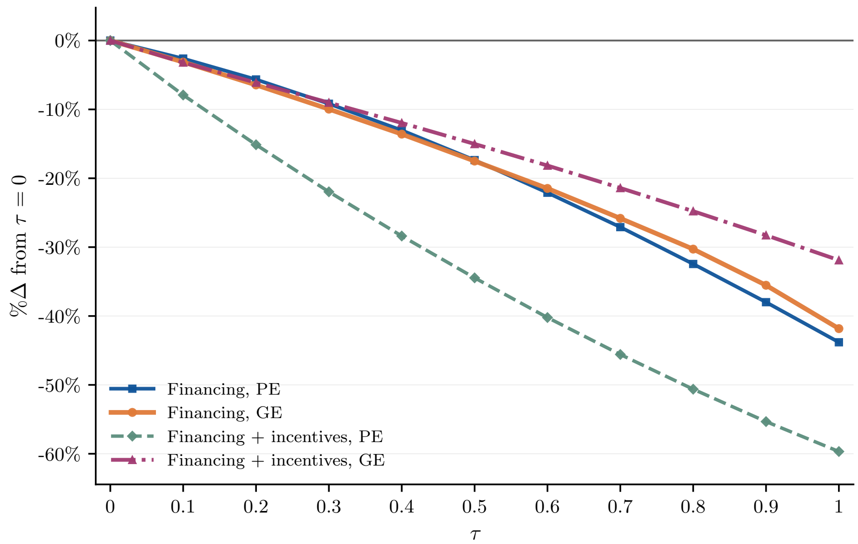
(b) Share constrained



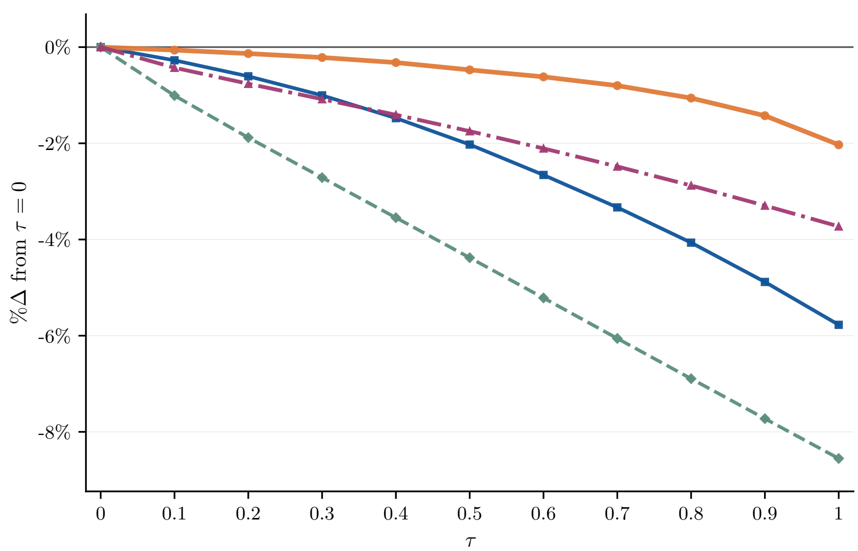
Notes: This figure reports the effects of the internal-finance wedge on firms' equity and financing constraints. Panel (a) reports the percentage change in aggregate equity relative to the baseline economy, while Panel (b) reports the percentage change in the share of firms operating at the borrowing constraint.

Figure 7. Impact on Markup Rents and Average Markups

(a) Markup rents



(b) Average markups



Notes: This figure compares the response of aggregate markup rents with the response of average markups. Panel (a) reports the percentage change in aggregate markup rents relative to the baseline economy. Panel (b) reports the percentage change in the average revenue-weighted markup relative to the baseline economy. The corresponding unweighted average markup is reported in Figure C13.

The comparison between partial- and general-equilibrium outcomes highlights the role of aggregate adjustment. For innovation, the differences are relatively modest, indicating that the effect of the wedge primarily operates through firm-level decisions. For output, however, equilibrium adjustment substantially dampens the contraction. In particular, the output loss falls from 10.8% to 4.5% when prices are allowed to adjust. Thus, while the incentive channel is quantitatively important, general-equilibrium forces partially offset its aggregate consequences. The financing channel, by contrast, remains economically meaningful under both partial and general equilibrium.

Contrasting these responses with those obtained when only financing capacity is affected allows the financing and incentive channels to be separately quantified, as summarized in Table 14. At $\tau = 0.1$, the financing channel accounts for 11% of the decline in R&D participation and roughly 31% of the output decline in partial equilibrium. The corresponding general-equilibrium shares are larger, 13% for R&D participation and 74% for output, because equilibrium adjustments dampen the additional contraction generated by firms' endogenous operating responses. As the wedge increases, the financing channel accounts for a larger share of the total response because general-equilibrium adjustment dampens the incentive-driven response more strongly than the finance-driven response.

The response of equilibrium factor prices helps clarify why the partial- and general-equilibrium effects differ. Figure 5 reports the responses of the interest rate and wage to the internal-finance wedge. Since the wedge reduces firms' ability to accumulate equity, firms operate at a smaller scale, placing downward pressure on labor demand. At $\tau = 0.1$, the wage declines by 3.0% when only the financing channel is active and by 6.7% when firms also adjust their operating decisions. At the same time, the equilibrium interest rate increases by 23.4% and 3.5%, respectively, as a higher interest rate clears the capital market when aggregate equity falls. These price adjustments affect the two channels differently. The incentive channel depends on current factor prices: lower wages reduce the cost of production and innovation, while higher interest rates raise the cost of capital. The financing channel, instead, depends on the stock of accumulated equity. Although the higher interest rate raises the return on retained funds, it does not restore the equity removed by the wedge. The increase in the return on equity is therefore insufficient to offset the decline in the stock of equity. As a result, general-equilibrium adjustment is more effective at dampening the incentive-driven response than the finance-driven response.

Figure 6 sheds further light on the mechanism underlying the financing channel. Panel (a) shows that the internal-finance wedge reduces aggregate equity in all counterfactuals, confirming that restricting the retention of markup rents weakens firms' internal financing capacity. Panel (b) shows the corresponding response of borrowing constraints. The increase in the share of constrained firms is largest when only the financing channel is active. In this case, firms continue to operate according to their baseline production and innovation policies, so financing needs remain largely unchanged while equity declines. As a result, more firms are pushed against the borrowing constraint. By contrast, when firms are allowed to adjust their operating decisions, they reduce production, capital demand, and innovation in response to the wedge. Financing needs therefore decline alongside financing capacity, muting the increase in the fraction of constrained firms.

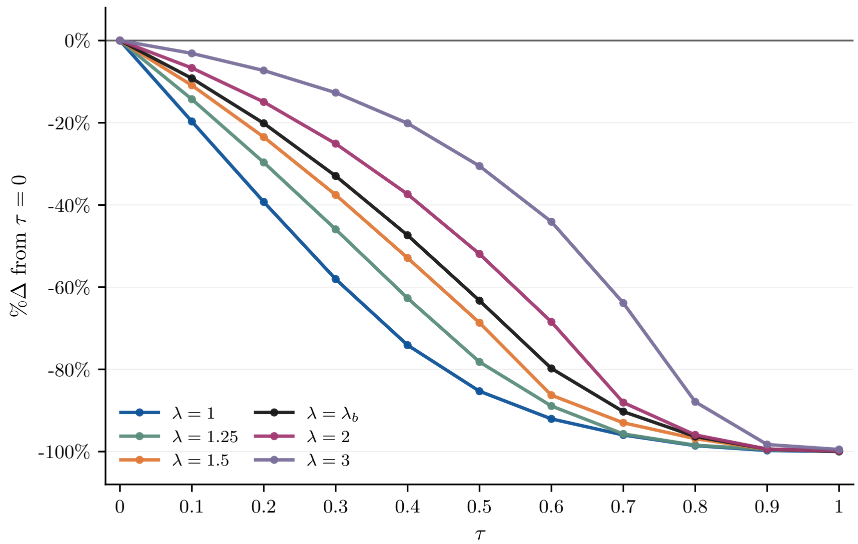
It is also important to ask whether restricting retained markup rents also reduces market power in the economy. The effect is ambiguous ex ante. On the one hand, limiting retained rents weakens the ability of large firms to finance future expansion, potentially reducing concentration and lowering markups. On the other hand, retained earnings are particularly important for financially constrained firms seeking to expand and compete with incumbents. Restricting internal funds may therefore hinder competitive pressure and preserve existing market power. Figure 7 evaluates these competing forces. The results indicate that restricting the accumulation of markup rents has a much larger effect on aggregate rents than on markups themselves. At $\tau = 0.1$, aggregate markup rents fall by 2.6% in partial equilibrium exercise and by 3.1% in general equilibrium when only the financing channel is active. When firms also adjust their operating decisions, markup rents fall by 7.9% in partial equilibrium and by 3.2% in general equilibrium. By contrast, the revenue-weighted average markup changes only modestly: at $\tau = 0.1$, it falls by 0.3% in the financing partial-equilibrium exercise, by 0.1% in the corresponding general-equilibrium exercise, by 1.0% in the financing-and-incentives partial-equilibrium exercise, and by 0.4% in general equilibrium. Thus, reducing firms' ability to retain markup rents lowers the aggregate flow of rents generated by market power, but does not substantially compress the markups that firms charge. The comparison between revenue-weighted and unweighted markups provides additional insight into the composition of the adjustment. While revenue-weighted markups decline slightly, the corresponding unweighted markups, reported in Figure C13, exhibit somewhat larger movements and even increases in general equilibrium. This pattern indicates that the wedge mainly shifts the scale and composition of activity across firms, with most of the markup changes occurring among firms that account for a small share of aggregate sales.

5.4 How Borrowing Constraints Influence the Markup-Rent Financing Channel

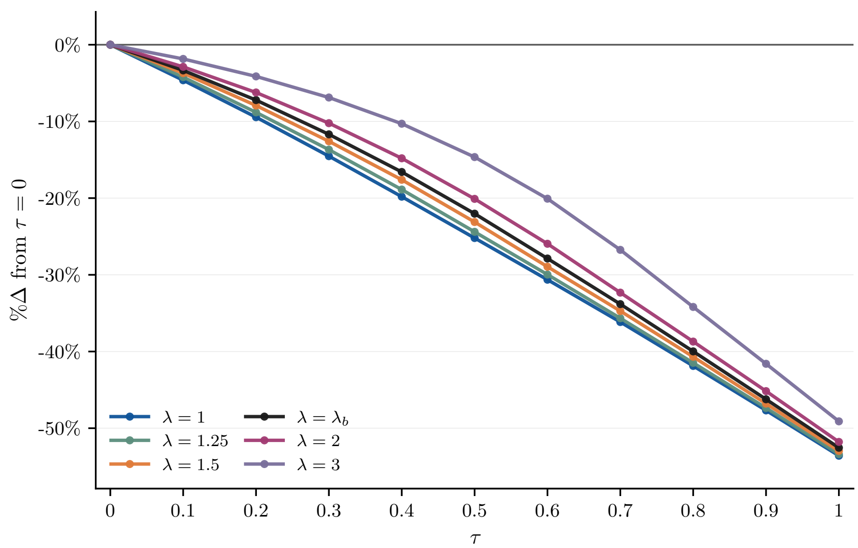
The previous section showed that markup rents affect aggregate innovation and output through a distinct financing channel: by increasing retained earnings, markup rents help firms accumulate equity and relax future borrowing constraints. A further check on this mechanism is to examine whether the effect is stronger when firms rely more heavily on internal funds. I therefore study how the response to the internal-finance wedge varies with the tightness of borrowing constraints. If the mechanism operates through firms' balance sheets, then restricting the accumulation of markup rents should have larger effects when borrowing constraints are tighter and external finance is harder to access. Conversely, in economies with looser borrowing constraints, firms should be better able to substitute external finance for retained earnings, weakening the markup-rent financing channel.

Figure 8. Aggregate Innovation and Output under Alternative Borrowing Constraints

(a) R&D participation



(b) Output



Notes: This figure reports the effects of the internal-finance wedge under alternative borrowing constraints. Outcomes are reported as percentage deviations from the corresponding economy with $\tau = 0$. The exercise is conducted in partial equilibrium. λ_b corresponds to the value of λ targeted in the baseline calibration. The response of aggregate R&D labor is qualitatively similar and is reported in Figure C14.

To implement this check, I repeat the internal-finance wedge exercise for alternative values of the leverage parameter λ, which governs how much capital a firm can operate for a given level of equity. Lower values of λ correspond to tighter borrowing constraints and therefore greater dependence on retained earnings. I conduct the analysis in partial equilibrium, holding aggregate prices fixed at the levels of the baseline unconstrained economy. I also hold firms' operating policies fixed, so technology choice, factor demands, pricing, output, and innovation decisions do not directly respond to τ. The wedge affects only the dynamic accumulation of equity. As a result, comparing across values of λ shows how the importance of the markup-rent financing channel varies with access to external funds.

Figure 8 shows that the effects of the internal-finance wedge are substantially larger when borrowing constraints are tighter. For every value of τ, restricting the accumulation of markup rents produces larger declines in innovation and output in economies with lower values of λ. Conversely, as borrowing constraints become less severe, the response of both innovation and output to the wedge becomes progressively weaker. When firms can borrow more easily, the loss of internal funds can be partially offset through external finance, reducing the importance of the markup-rent financing channel. However, the effects remain even in relatively unconstrained economies, indicating that internal funds are not perfectly substitutable with external finance. Nevertheless, the relationship between λ and the response to the wedge provides support that the mechanism operates through borrowing constraints. If markup rents affected innovation solely through contemporaneous profitability, changing the tightness of borrowing constraints would have little effect on the estimated responses. Instead, the importance of the wedge declines as access to external finance improves, as predicted by the financing-channel interpretation.

Overall, these results show that financial frictions shape the strength of the markup-rent financing channel. Markup rents matter for innovation at all levels of λ, but they matter most when borrowing constraints are tight. When external finance is limited, retained markup-rents provide an important source of internal funds that allows firms to accumulate equity and finance R&D. As borrowing constraints relax, firms become less dependent on retained markup-rents, and the innovation loss from partially restricting those rents becomes smaller. This supports the interpretation of market power as an imperfect substitute for missing financial markets.

6 Conclusion

This paper studied how market power affects innovation when firms face financial frictions. Using administrative data covering the population of non-financial firms in Portugal, I documented that innovation is concentrated among firms that are larger, more productive, charge higher markups, and rely more on equity finance. I also showed that higher markups are associated with faster equity accumulation, consistent with the interpretation that markup rents strengthen firms' internal financing capacity. Motivated by these facts, I developed and quantified a general equilibrium model with heterogeneous firms, endogenous innovation, financial frictions, and variable markups. Viewed through the lens of the quantified model, market power influences innovation not only by shaping firms' incentives, but also by affecting their ability to finance future R&D. Quantitatively, the financing role of market power accounted for a meaningful share of the response of aggregate innovation and output to changes in retained markup rents. Counterfactual exercises further showed that the importance of this channel increased with the severity of borrowing constraints. More broadly, the findings suggest that the effects of competition policy depend on the financial environment in which firms operate. In economies where firms can readily finance innovation through capital markets, reducing market power is less likely to impede innovative activity. In economies where external finance remains limited, however, markup rents may partially substitute for missing financial markets by providing the internal resources needed to fund innovation and growth.

References

Acemoglu, D., Akcigit, U., Bloom, N., and Kerr, W. (2018). Innovation, reallocation, and growth. *American Economic Review*, 108(11):3450–3491.

Ackerberg, D. A., Caves, K., and Frazer, G. (2015). Identification Properties of Recent Production Function Estimators. *Econometrica*, 83(6).

Aghion, P., Bergeaud, A., Boppart, T., Klenow, P. J., and Li, H. (2023). A theory of falling growth and rising rents. *The Review of Economic Studies*, 90(6):2675–2702.

Aghion, P., Bergeaud, A., Boppart, T., Klenow, P. J., and Li, H. (2025). Good rents versus bad rents: R&d misallocation and growth. NBER Working Paper 34190, National Bureau of Economic Research.

Aghion, P., Bloom, N., Blundell, R., Griffith, R., and Howitt, P. (2005). Competition and innovation: An inverted-u relationship. *The Quarterly Journal of Economics*, 120(2):701–728.

Aghion, P. and Howitt, P. (1992). A model of growth through creative destruction. *Econometrica*, 60(2):323–351.

Arrow, K. J. (1962). Economic welfare and the allocation of resources for invention. In *The Rate and Direction of Inventive Activity: Economic and Social Factors*, pages 609–626. National Bureau of Economic Research, Princeton, NJ.

Autor, D., Dorn, D., Katz, L. F., Patterson, C., and Van Reenen, J. (2020). The fall of the labor share and the rise of superstar firms. *The Quarterly Journal of Economics*, 135(2):645–709.

Banco de Portugal Microdata Research Laboratory (BPLIM) (2025). Central Balance Sheet Harmonized Panel.

Bloom, N., Jones, C. I., Van Reenen, J., and Webb, M. (2020). Are ideas getting harder to find? *American Economic Review*, 110(4):1104–44.

Boar, C. and Midrigan, V. (2025). Markups and inequality. *Review of Economic Studies*, 92(5):2828–2860.

Brown, J. R., Fazzari, S. M., and Petersen, B. C. (2009). Financing innovation and growth: Cash flow, external equity, and the 1990s r&d boom. *Journal of Finance*, 64(1):151–185.

Buera, F. J., Kaboski, J. P., and Shin, Y. (2011). Finance and development: A tale of two sectors. *American Economic Review*, 101(5):1964–2002.

Buera, F. J. and Shin, Y. (2013). Financial frictions and the persistence of history: A quantitative exploration. *Journal of Political Economy*, 121(2).

Cagetti, M. and De Nardi, M. (2006). Entrepreneurship, Frictions, and Wealth. *Journal of Political Economy*, 114(5).

Caggese, A. (2019). Financing constraints, radical versus incremental innovation, and aggregate productivity. *American Economic Journal: Macroeconomics*, 11(2):238–272.

Crouzet, N. and Eberly, J. C. (2018). Intangibles, investment, and efficiency. *AEA Papers and Proceedings*, 108:426–431.

Crouzet, N. and Eberly, J. C. (2019). Understanding weak capital investment: The role of market concentration and intangibles. NBER Working Paper 25869, National Bureau of Economic Research.

Crouzet, N., Eberly, J. C., Eisfeldt, A. L., and Papanikolaou, D. (2022). The economics of intangible capital. *Journal of Economic Perspectives*, 36(3):29–52.

De Loecker, J., Eeckhout, J., and Unger, G. (2020). The rise of market power and the macroeconomic implications. *The Quarterly Journal of Economics*, 135(2):561–644.

De Loecker, J. and Warzynski, F. (2012). Markups and Firm-Level Export Status. *American Economic Review*, 102(6).

Edmond, C., Midrigan, V., and Xu, D. Y. (2023). How costly are markups? *Journal of Political Economy*, 131(7):1619–1675.

Gilbert, R. J. and Newbery, D. M. G. (1982). Preemptive patenting and the persistence of monopoly. *American Economic Review*, 72(3):514–526.

Gopinath, G., Kalemli-Özcan, , Karabarbounis, L., and Villegas-Sanchez, C. (2017). Capital Allocation and Productivity in South Europe. *Quarterly Journal of Economics*, 132(4).

Hall, B. H. and Lerner, J. (2010). The financing of r&d and innovation. In Hall, B. H. and Rosenberg, N., editors, *Handbook of the Economics of Innovation*, volume 1, chapter 14, pages 609–639. Elsevier.

Hall, R. E. (1988). The Relation between Price and Marginal Cost in U.S. Industry. *Journal of Political Economy*, 96(5).

Hsieh, C.-T. and Klenow, P. J. (2009). Misallocation and manufacturing TFP in China and India. *Quarterly Journal of Economics*, 124(4):1403–1448.

Jones, B. F. (2009). The burden of knowledge and the "death of the renaissance man": Is innovation getting harder? *The Review of Economic Studies*, 76(1):283–317.

Kalemli-Özcan, Ş., Sørensen, B. E., Villegas-Sanchez, C., Volosovych, V., and Yeşiltaş, S. (2024). How to construct nationally representative firm-level data from the orbis global database: New facts on smes and aggregate implications for industry concentration. *American Economic Journal: Macroeconomics*, 16(2):353–374.

Klenow, P. J. and Willis, J. L. (2016). Real Rigidities and Nominal Price Changes. *Economica*, 83(331).

Klette, T. J. and Kortum, S. (2004). Innovating firms and aggregate innovation. *Journal of Political Economy*, 112(5):986–1018.

Levinsohn, J. and Petrin, A. (2003). Estimating production functions using inputs to control for unobservables. *Review of Economic Studies*, 70(2).

Lian, C. and Ma, Y. (2021). Anatomy of corporate borrowing constraints. *The Quarterly Journal of Economics*, 136(1):229–291.

Liang, Y. (2023). Misallocation and markups: Evidence from indian manufacturing. *Review of Economic Dynamics*, 51:161–176.

Midrigan, V. and Xu, D. Y. (2014). Finance and misallocation: Evidence from plant-level data. *American Economic Review*, 104(2):422–458.

Moll, B. (2014). Productivity losses from financial frictions: Can self-financing undo capital misallocation? *American Economic Review*, 104(10):3186–3221.

Nanda, R. and Kerr, W. R. (2015). Financing innovation. *Annual Review of Financial Economics*, 7(1):445–462.

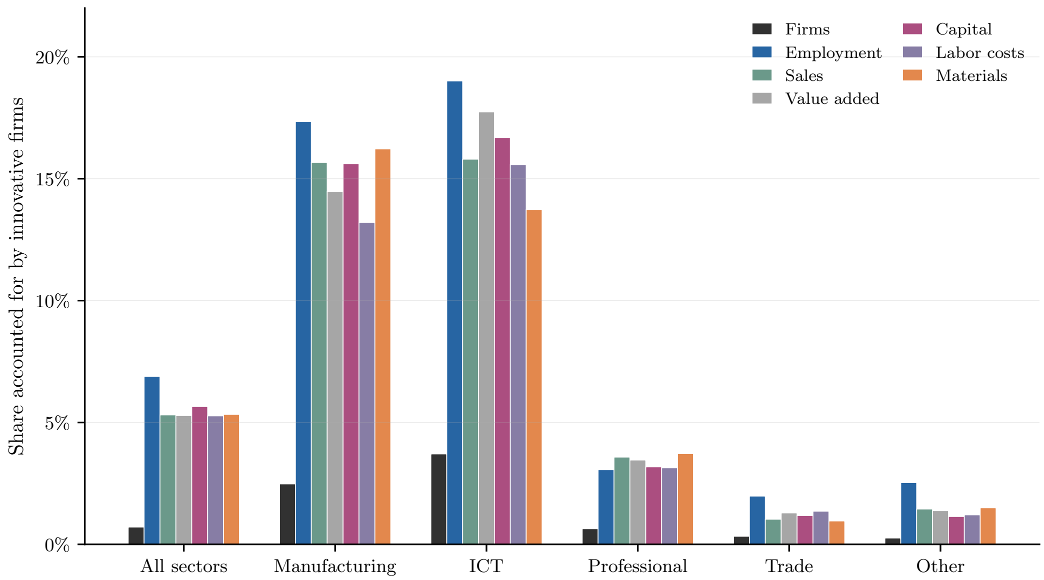
Peters, R. H. and Taylor, L. A. (2017). Intangible capital and the investment-q relation. *Journal of Financial Economics*, 123(2):251–272.

Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy*, 98(5):S71–S102.

Schumpeter, J. A. (1942). *Capitalism, Socialism and Democracy*. Harper & Brothers, New York.

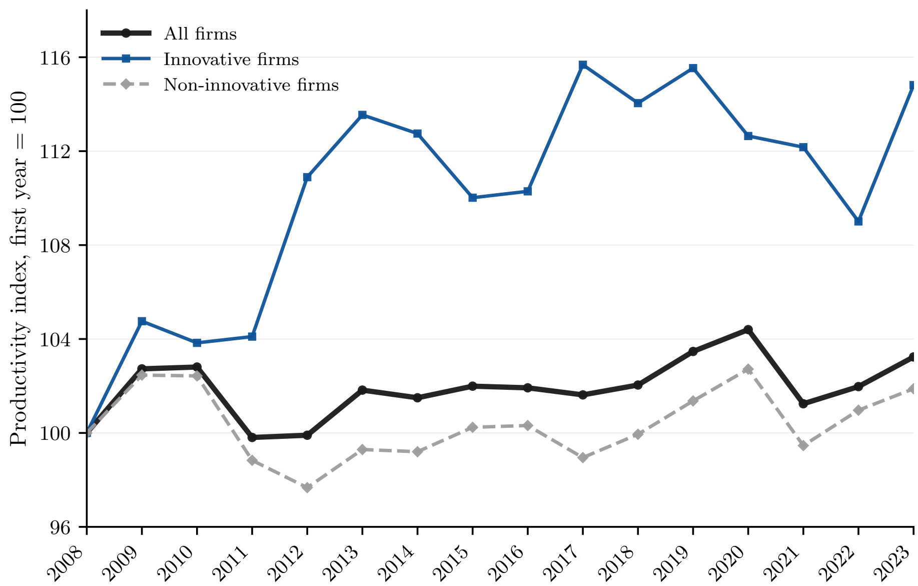
A Additional Figures

Figure A9. Aggregate Importance of Innovative Firms



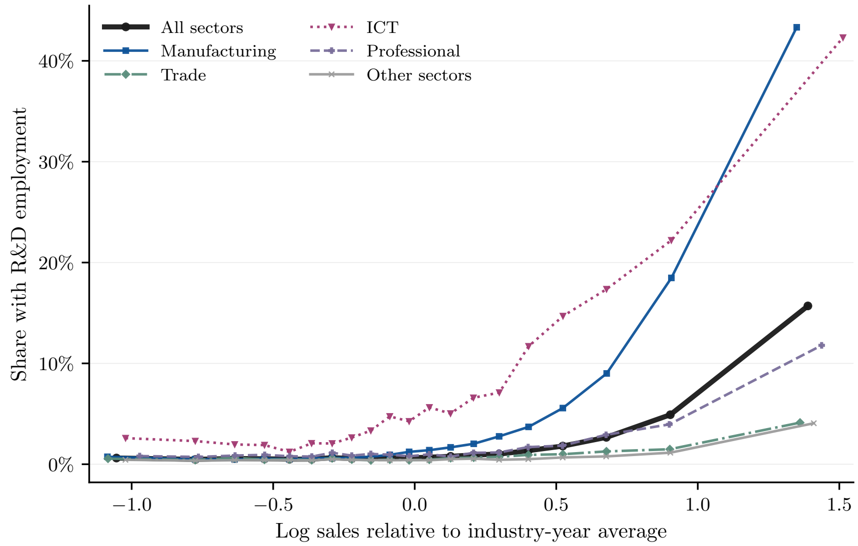
Notes: The figure reports the aggregate shares accounted for by firms with positive R&D employment across sector groups. The bars show the share of firms, employment, sales, value added, capital, labor costs, and materials expenditures associated with innovative firms. Value added is defined as sales minus materials expenditures. Shares are computed separately by year and then averaged over the 2006–2023 period. Innovative firms are defined as firms with positive R&D employment. Sector groups correspond to NACE Rev. 2 sections: Manufacturing (C), Wholesale and retail trade; repair of motor vehicles and motorcycles (G), Information and communication (J), Professional, scientific and technical activities (M), and a residual category including all remaining sectors. The "All sectors" category refers to the full sample of firms across all sectors.

Figure A10. Aggregate Productivity Growth

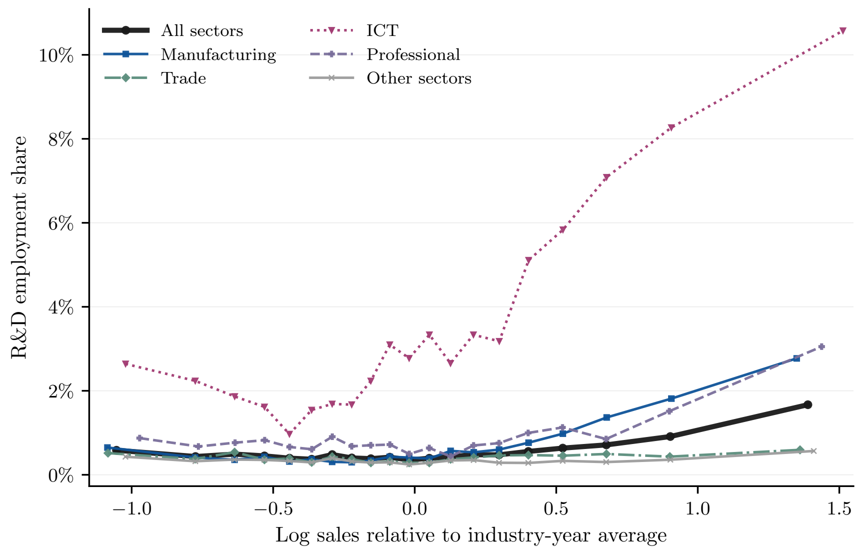


Notes: The figure plots gross-output-weighted productivity indices for innovative firms, non-innovative firms, and the aggregate economy over the 2008–2023 period. Innovative firms are defined as firms with positive R&D employment. Productivity is measured using the log productivity estimates recovered from the Cobb–Douglas production function estimation. Firm-level productivity is aggregated using gross-output weights, where gross output is defined as sales net of material expenditures. Aggregate productivity is computed as the weighted average of productivity across innovative and non-innovative firms using each group's share in total gross output. All series are normalized to 100 in 2008. Although the sample begins in 2006, the series starts in 2008 because the production function estimation requires lagged information to recover firm-level productivity estimates.

Figure A11. R&D Employment and Firm Size by Sector



(a) Share with R&D employment



(b) R&D employment share

Notes: The figure plots R&D activity by relative firm size separately for each sector group. Firm size is measured as log sales relative to the average log sales within the same 2-digit industry-year cell. Firms are first grouped into ventiles based on relative firm size. Within each ventile, Panel (a) reports the share of firms with positive R&D employment, while Panel (b) reports the average firm-level ratio of R&D employment to total employment. The horizontal axis is expressed in $\log_{10}$ terms, so equal distances correspond to proportional differences in sales relative to the industry-year average. For example, a value of 1 corresponds to firms with sales 10 times larger than the industry-year average, while a value of -1 corresponds to firms with sales one-tenth as large. Sector groups correspond to NACE Rev. 2 sections: Manufacturing (C), Wholesale and retail trade; repair of motor vehicles and motorcycles (G), Information and communication (J), Professional, scientific and technical activities (M), and a residual category including all remaining sectors.

B Additional Regression Results

Table B15. R&D Employment, Productivity, and Markups with Industry-by-Year Fixed Effects

	$\ln(\text{Productivity}_{it})$		$\ln(\text{Markup}_{it})$	
	(1)	(2)	(3)	(4)
$\ln(\text{R\&D emp}_{it})$	0.0187**	0.0175*	0.0165**	0.0155*
	(0.0087)	(0.0090)	(0.0081)	(0.0085)
Industry FE	Y	-	Y	-
Year FE	Y	-	Y	-
Industry $\times$ Year FE	-	Y	-	Y
Firm controls	Y	Y	Y	Y
Overhead ratio	Y	Y	Y	Y
Observations	13,478	13,365	13,478	13,365
R-squared	0.8358	0.8436	0.4288	0.4581

Notes: Columns (1) and (2) use the log of firm-level productivity estimated following De Loecker and Warzynski (2012) under a Cobb–Douglas production function as the dependent variable. Columns (3) and (4) use the log of firm-level markups estimated following De Loecker and Warzynski (2012). Columns (1) and (3) correspond to the baseline specifications from Table 6 with separate 2-digit industry and year fixed effects. Columns (2) and (4) replace these with 2-digit industry-by-year fixed effects. Firm-level controls include firm age and size (measured by the log of total assets), exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. All specifications additionally control for overhead relative to total expenses as a proxy for fixed operating costs. Robust standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B16. Innovation Activity and Access to External Finance

	$\mathbb{1}(\text{Total debt}_{it} > 0)$			$\mathbb{1}(\text{Long-term debt}_{it} > 0)$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}(\text{R\&D emp}_{it} > 0)$	-0.0467***	-0.0431***	-0.0401***	-0.0166**	-0.0134*	-0.0087
	(0.0103)	(0.0090)	(0.0094)	(0.0076)	(0.0077)	(0.0076)
Industry FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y	Y	Y
Cash ratio	-	Y	Y	-	Y	Y
Collateral ratio	-	-	Y	-	-	Y
Observations	1,074,551	1,074,551	1,074,551	1,074,551	1,074,551	1,074,551
R-squared	0.1378	0.1789	0.1853	0.1262	0.1493	0.1603

Notes: Columns (1)–(3) use an indicator equal to one if the firm holds positive total debt as the dependent variable. Columns (4)–(6) use an indicator equal to one if the firm holds positive long-term debt. $\mathbb{1}(\text{R\&D emp} > 0)$ is an indicator equal to one if the firm employs R&D workers. All specifications include 2-digit industry and year fixed effects. Firm-level controls include firm age and size (measured by the log of total assets), exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. The cash ratio is defined as cash and cash equivalents relative to total assets. The collateral ratio is defined as tangible fixed assets relative to total assets. Columns (3) and (6) additionally control for both the cash ratio and the collateral ratio. Robust standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B17. Innovation Activity and Debt Maturity Structure

	Short-term leverage ratio$_{it}$		Long-term leverage ratio$_{it}$	
	(1)	(2)	(3)	(4)
$\mathbb{1}(\text{R\&D emp}_{it}) > 0)$	-0.0054***	-0.0044***	-0.0130***	-0.0080***
	(0.0015)	(0.0016)	(0.0031)	(0.0028)
Industry FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y
Cash ratio	-	Y	-	Y
Collateral ratio	-	Y	-	Y
Observations	1,074,551	1,074,551	1,074,551	1,074,551
R-squared	0.1232	0.1370	0.0459	0.1225

Notes: Columns (1) and (2) use the short-term leverage ratio, defined as short-term debt relative to total assets, as the dependent variable. Columns (3) and (4) use the long-term leverage ratio, defined as long-term debt relative to total assets. $\mathbb{1}(\text{R\&D emp} > 0)$ is an indicator equal to one if the firm employs R&D workers. All specifications include 2-digit industry and year fixed effects. Firm-level controls include firm age and size (measured by the log of total assets), exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. Columns (2) and (4) additionally control for the cash ratio, defined as cash and cash equivalents relative to total assets, and the collateral ratio, defined as tangible fixed assets relative to total assets. Robust standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B18. Innovation Activity and Cost of Debt

	Cost of debt_{it}		
	(1)	(2)	(3)
$\mathbb{1}(\text{R\&D emp}_{it} > 0)$	-0.0027*	-0.0023*	-0.0023*
	(0.0015)	(0.0014)	(0.0014)
Industry FE	Y	Y	Y
Year FE	Y	Y	Y
Firm controls	Y	Y	Y
Collateral ratio	-	Y	Y
Leverage ratio	-	Y	Y
Maturity structure	-	-	Y
Observations	831,453	831,453	831,453
R-squared	0.0634	0.1145	0.1161

Notes: Columns (1)–(3) use the cost of debt, defined as interest expenses relative to total debt, as the dependent variable. $\mathbb{1}(\text{R\&D emp} > 0)$ is an indicator equal to one if the firm employs R&D workers. All specifications include 2-digit industry and year fixed effects. Firm-level controls include firm age and size, measured by the log of total assets, exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. Columns (2) and (3) additionally control for the collateral ratio, defined as tangible fixed assets relative to total assets, and the leverage ratio, defined as total debt relative to total assets. Column (3) further controls for debt maturity structure, measured as long-term debt relative to total debt. Robust standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B19. Market Power and Equity Accumulation Controlling for Investment

	$\Delta \ln(\text{Equity}_{it})$			
	All firms		*Innovative firms*	
	(1)	(2)	(3)	(4)
$\ln(\text{Markup}_{it})$	0.0920***	0.0947***	0.0757***	0.0759***
	(0.0123)	(0.0123)	(0.0124)	(0.0115)
Industry FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y
Collateral ratio	Y	Y	Y	Y
Leverage ratio	Y	Y	Y	Y
Investment rate	-	Y	-	Y
Observations	1,550,506	1,550,506	13,478	13,478
R-squared	0.0451	0.0531	0.0472	0.0721

Notes: The dependent variable is the growth rate of firm equity, measured as the first difference of log equity. Firm-level markups are estimated following De Loecker and Warzynski (2012) under a Cobb–Douglas production function specification. Innovative firms are defined as firms employing R&D workers. All specifications include 2-digit industry and year fixed effects. Firm-level controls include firm age and size (measured by the log of total assets), exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. All specifications additionally control for the collateral ratio, defined as the ratio of tangible assets to total fixed assets, and the leverage ratio, defined as total debt relative to total assets. Columns (2) and (4) additionally control for the investment rate, measured as the first difference of log capital. Robust standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

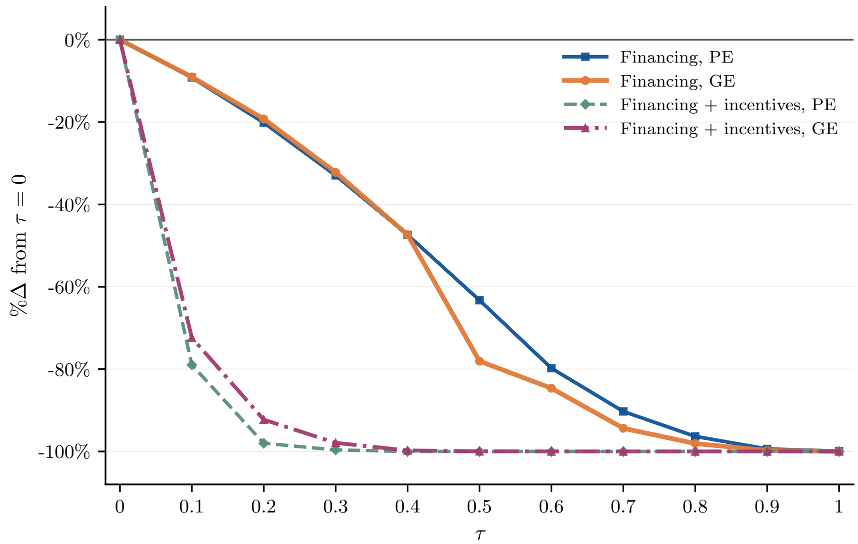
Table B20. Market Power and Equity Accumulation with Industry-by-Year and Firm Fixed Effects

	$\Delta\ln(\text{Equity}_{it})$					
	All firms			*Innovative firms*		
	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(\text{Markup}_{it})$	0.0920***	0.0917***	0.3018***	0.0682***	0.0673***	0.1847***
	(0.0123)	(0.0115)	(0.0357)	(0.0119)	(0.0116)	(0.0373)
Industry FE	Y	-	-	Y	-	-
Year FE	Y	-	-	Y	-	-
Industry $\times$ Year FE	-	Y	Y	-	Y	Y
Firm FE	-	-	Y	-	-	Y
Firm controls	Y	Y	Y	Y	Y	Y
Collateral ratio	Y	Y	Y	Y	Y	Y
Leverage ratio	Y	Y	Y	Y	Y	Y
Observations	1,550,506	1,550,500	1,504,447	13,478	13,365	11,639
R-squared	0.0451	0.0520	0.2694	0.0472	0.1102	0.3760

Notes: The dependent variable is the growth rate of firm equity, measured as the first difference of log equity. Firm-level markups are estimated following De Loecker and Warzynski (2012) under a Cobb–Douglas production function specification. Innovative firms are defined as firms employing R&D workers. Columns (1) and (4) reproduce the baseline specifications from Table 9. Columns (2) and (5) replace separate industry and year fixed effects with 2-digit industry-by-year fixed effects. Columns (3) and (6) additionally include firm fixed effects. Firm-level controls include firm age and size (measured by the log of total assets), exporter status, legal form indicators, district fixed effects, a dummy for single-owner firms, and an indicator for multi-establishment firms. In the firm fixed effects specifications, time-invariant controls are absorbed by the fixed effects and omitted to avoid collinearity. All specifications additionally control for the collateral ratio and leverage ratio. Robust standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

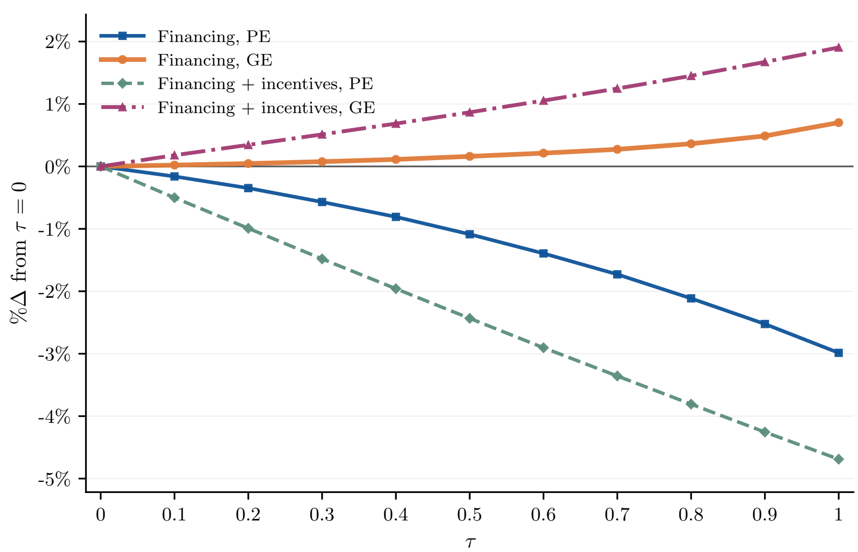
C Additional Model Results

Figure C12. Impact on Aggregate R&D Labor



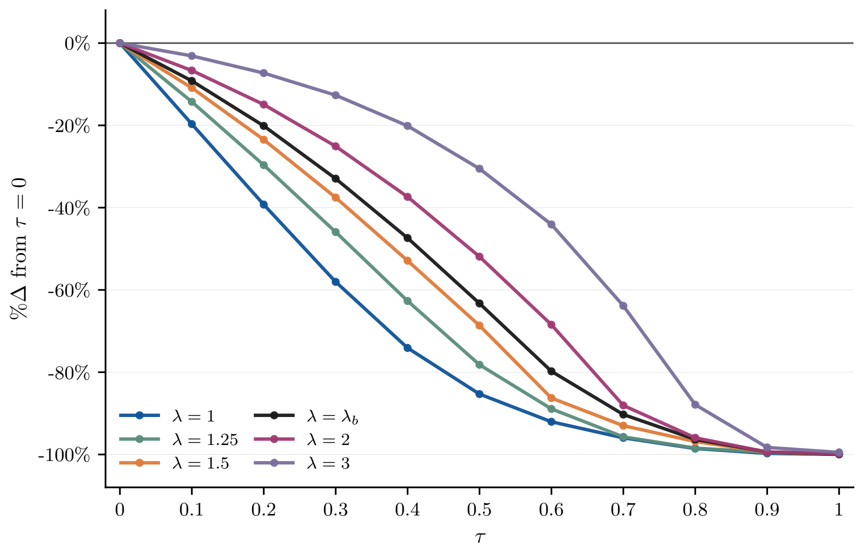
Notes: This figure reports the effect of the markup-rent financing wedge on aggregate R&D labor. Figure 4a shows the participation margin.

Figure C13. Unweighted Average Markups



Notes: This figure reports the percentage change in the unweighted average markup relative to the baseline economy. The corresponding revenue-weighted average markups are shown in Figure 7b.

Figure C14. Aggregate R&D Labor under Alternative Borrowing Constraints



Notes: This figure reports the effects of the internal-finance wedge on aggregate R&D labor under alternative borrowing constraints. Outcomes are reported as percentage deviations from the corresponding economy with $\tau = 0$. The exercise is conducted in partial equilibrium. λ_b corresponds to the value of λ targeted in the baseline calibration. Figure 8a shows the participation margin.